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1. Introduction

Our company's goal is to let customers have the most comfortable user experience. The development of our intelligent warehousing system will bring users a satisfactory experience. We are mainly engaged in the development of intelligent warehousing products, and our products are highly praised in the industry now. Different from the ordinary intelligent freight system, our products will provide users with the best service and the most accurate goods delivery. Our aim is to let every customer be satisfied with our products.



In terms of hardware, the warehouse has realized unmanned and automation, and the intelligent car is used to make all functions be achieved more effectively, such as tracing the warehouse, detecting dangerous situations, etc. The intelligent car can also automatically identify materials and easily get access to them according to computer instructions, improving the efficiency and saving manpower.

In terms of software, the website has realized displaying data intelligently. Administrators and workers can directly log in the intelligent warehousing website to realize online goods query and management. The website system is flexibly configured to solve the problems of multiple products and inventory confusion.

Product advantages

Our intelligent freight management system is simple to operate and highly integrated. Compared with other intelligent freight management systems, our intelligent freight management system has more perfect functions. Users only need to log in to our intelligent freight management homepage to control the freight vehicles, view all the information of workers and administrators, and check the status of each warehouse. In addition, we also provide users with many optional operations to ensure that users can control all aspects of all links they want to know through our system.

Product development prospect

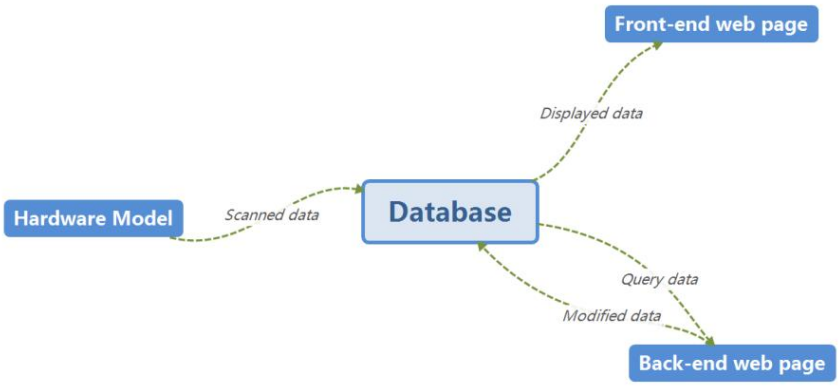
Nowadays, information science is increasingly developed in the world, 5G technology is gradually entering people's lives, and the interconnection of everything has become the trend of the times. The traditional freight transport system with cumbersome processes is gradually eliminated,

replaced by an intelligent freight transport system that is easy to operate and has powerful functions. Users can easily master and adjust the situation of each link in the transportation process through our intelligent freight management system, which is a convenience that traditional freight systems cannot bring. In addition, this industry is a new industry, and its potential has not been fully explored, so it has a very broad development prospect.

2. Function Design

2.1. Overall Design

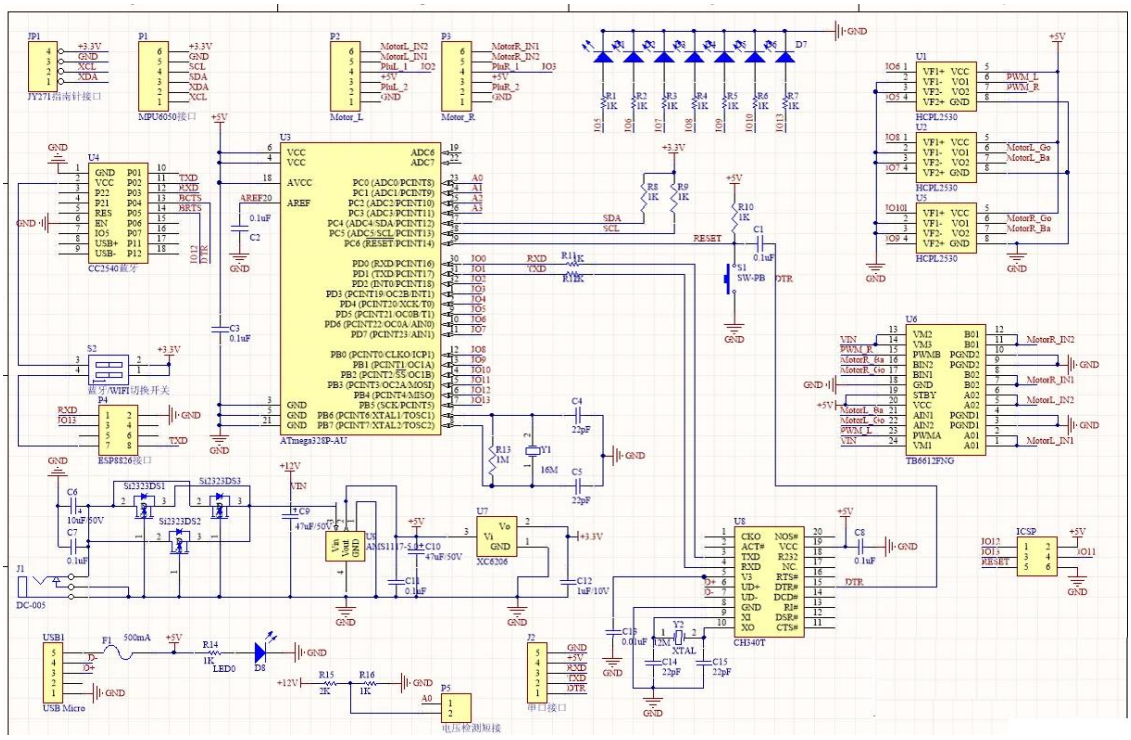
Our company co-designs and builds a warehouse monitoring system that uses automated robots to work on-site and send a status report of the warehouse to a web application for monitoring. At the same time, there are two web programs for workers to manage warehouse goods and for managers to manage workers. Members in our group from 3 different majors first implemented different functions at the same time through division of labor, and then integrated the results. At the beginning, the students of electrical engineering were responsible for the design and construction of the hardware model of the automatic robot, the students of Internet of Things were responsible for the construction of the hardware database interface, database and the client software of the workers, and the students of e-commerce were responsible for the database management and the construction of the management software. After we finished our assigned tasks, we put the different parts together. The connection mode is shown as follows. The data is transferred from the hardware model to the database, which provides the data to the back-end website and finally presents it on the web page.

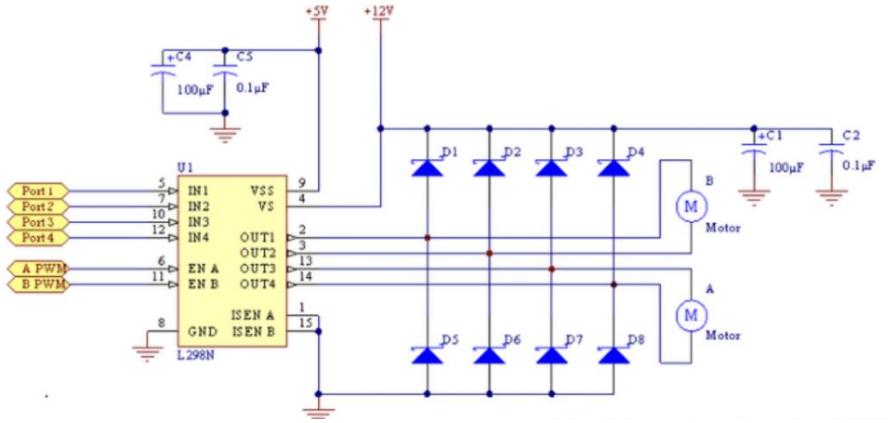


2.2. Respective Design

2.2.1. Hardware

(1) Circuit Diagram Design





List of The Main Hardware is as follows:

NO.	NAME	NUMBER
1	Arduino uno	1
2	Arduino mega	1
3	Arduino extended board	1
4	Temperature and humidity sensor	1
5	TCRT5000 Infrared sensor	3
6	The LCD screen	1
7	Flame sensor	1
8	ESP8266-01s Wi-Fi-module	1
9	Wi-Fi-module downloader	1
10	TCS3200 Color sensor	1

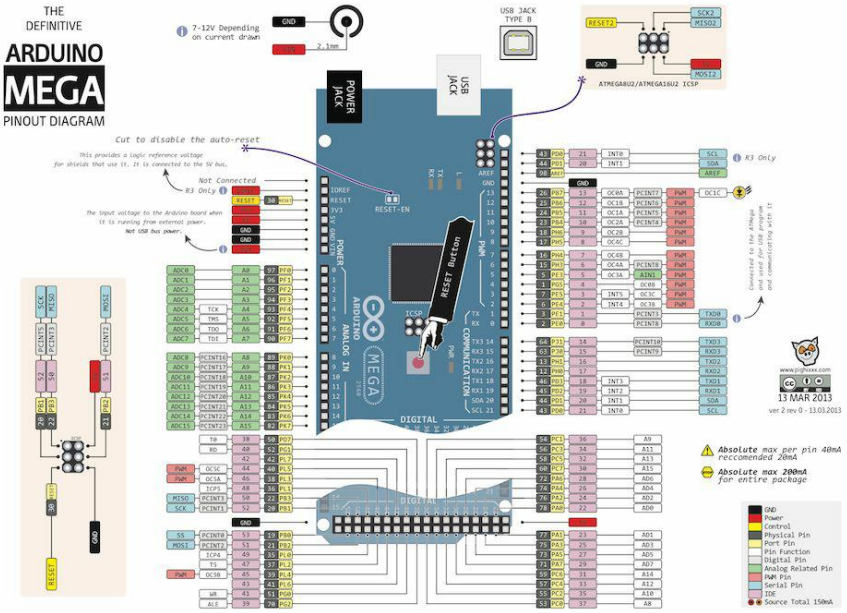
11	Photosensitive sensor	1
12	Buzzer	1
13	HC-SR04 Ultrasonic ranging module	1
14	L298N Drive motor control module	1
15	Motor	4
16	Infra-red obstacle avoidance module	1

(2) Introduction of Some Main Hardware

- Arduino Mega 2560 & Arduino Extended Board

The Arduino Mega 2560 is a core board with a USB interface. With 54 digital inputs and outputs, it is suitable for designs that require many IO interfaces. The core of the processor is the ATmega2560, which also has 54 digital inputs/outputs, 16 analogue inputs, 4 UART interfaces, a 16MHz crystal oscillator, a USB port, a power socket, an ICSP header and a reset button.

We also choose an Arduino extended board. It provides us with more interfaces for connecting more sensors and achieving more functions. It also reduces conflicts between the different ports. Therefore, it allows all components to work properly and data to be collected and uploaded successfully.



- TCRT5000 Infrared Sensor

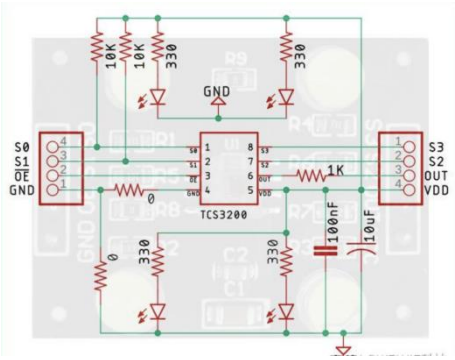
The TCRT5000 has an infrared transmitter tube and an infrared receiver tube. When the infrared signal from the transmitter tube is reflected and received by the receiver tube, the resistance of the receiver tube will change.



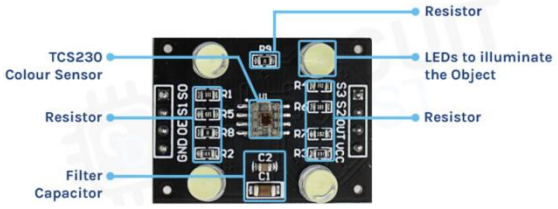
Go Straight Turn Right Turn Left

- Color Sensor

The TCS3200 has a programmable optical and electrical converter with a built-in oscillator that converts the light intensity into a square wave. The output value of the OUT pin of the TCS3200 is proportional to the frequency of the built-in oscillator and their ratio factor is set by its pins S0 and S1.



Circuit Diagram of TCS3200

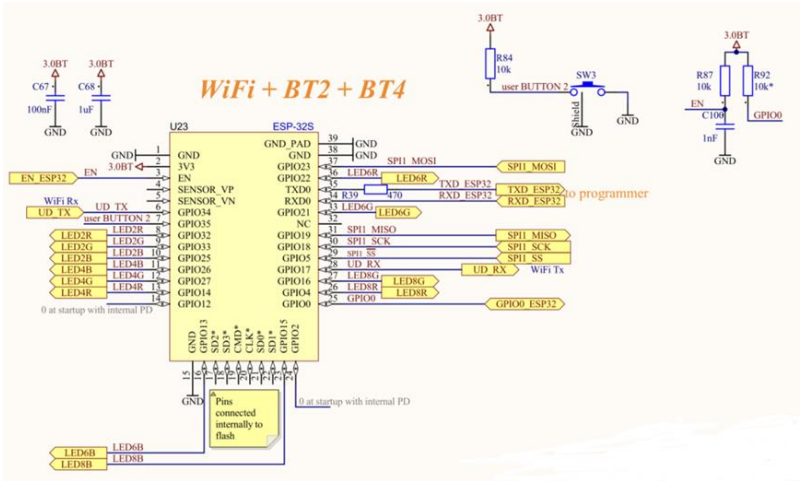


TCS3200 Component Parts

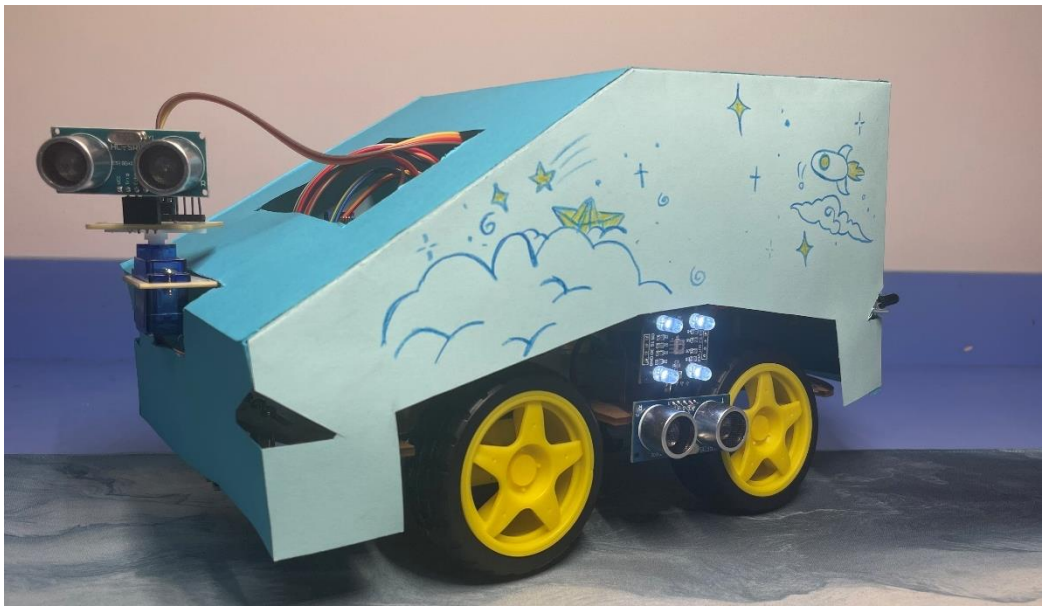
After testing, we found that the optimum distance between the object under testing and the TCS3200 is 1 cm. when detecting objects, it is important to avoid interference from external light, otherwise the results may be affected.

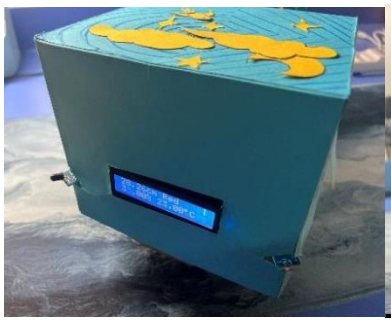
- ESP32 Wi-Fi Module

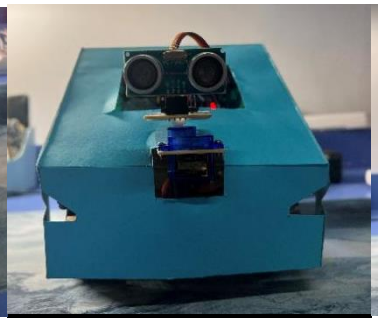
The ESP32 is an integrated 2.4 GHz Wi-Fi and Bluetooth dual-mode single-chip device using TSMC's ultra-low power 40nm process, offering ultra-high RF performance, stability, versatility, and reliability, as well as ultra-low power consumption to meet different power requirements for a wide range of application scenarios.

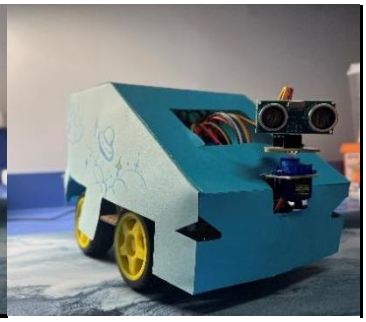


(3) Appearance









- The main body of our car has a rectangular shape. Several windows have been cut in many places to allow the sensors to be placed on the outside of the car which is better for measurement of the data we need. The compactness of the internal components and the appearance of the cart, which hides the tightly packed internal wiring, enhance the aesthetics and resistance to damage.
- The housing of the car is of a top-sleeve construction, making it easier to remove the exterior of the car for internal inspection and repair if problems arise inside.
- The parts left on the outside give the small car the more technological feeling. The use of environmentally friendly materials combined with the production of the cart's exterior not only saves resources, but also facilitates better mass production of this type of cart and helps to upgrade the warehouse management system.

(4) Systematic Design

- **Ideas for System Design**

  According to the requirement of warehouse managing, we think our car's system is expected

to be well-designed to satisfy the following demands:

✧ **Anomaly detection and reporting**

There may be some unexpected situations in the warehouse, such as intrusion by unknown persons, fire, and other abnormal situations. Therefore, there shall be real-time monitoring of abnormal information and timely alarms function so that the warehouse manager can handle the situation as quickly as possible.

✧ **Automate operations**

In unmanned warehouses, the trolley needs to be able to automatically cruise along the given path and automatically identify the current cargo-related information by detecting the color of the goods.

✧ **Environmental information monitoring**

The car needs to collect the environmental parameters of the warehouse (current temperature, humidity, etc.) and feedback to the computer in time to improve the preservation capacity of the goods.

✧ **Emergency response**

The travel route of the car may not be smooth and plain, and occasionally there will even be some obstacles on the path. Therefore, our car should respond to these unexpected situations on time.

The Main Function Introduction

As the hardware part of the warehouse supervision system, this car is designed with the following functions:

A. Environmental parameter acquisition

In modern warehouse management, the environmental information of the warehouse is changeable all the time. Among them, temperature and humidity are of great importance and play a role in the fire, mildew, and moisture resistance in the warehouse.

Therefore, we put a DHT11 temperature and humidity sensor on our car, which can measure the temperature and humidity of the warehouse environment continuously, in order to maintain the quality of stored goods and create an environment suitable for commodity storage.

The code is shown below:

```
void TandH()
{
  lcd.setCursor(0,1);            //第二行显示
  lcd.print(dht.readHumidity());       //输出字符 表示湿度
  lcd.setCursor(5,1);
  lcd.print("%");

  lcd.setCursor(7,1);
  lcd.print(dht.readTemperature());       //输出字符 表示温度
  lcd.setCursor(12,1);
  lcd.print(char(0xdf));
  lcd.setCursor(13,1);
  lcd.print("C");
}
```

B. Fire alarm

Obviously, if there is a fire in the warehouse, not only will it cause a significant loss of goods and damage to the environment, but also the security of nearby constructions and even the warehouse managers are threatened. Therefore, it is necessary to timely detect whether there is a fire in the warehouse or not.

In order to remedy the fire hazard in time, we put a flame sensor on our car, which can sense whether there is a flame around the car and if there is, light up to inform the manager, in order to enhance the fire safety management of the warehouse, respond to the fire situation promptly, and reduce losses as well.

The code is shown below:

```
void Bee_set()
{
  pinMode(DigitalInPin,INPUT);       //set the pins state
  pinMode(BuzzerPin,OUTPUT);
}

void Bee()
{
  digitalWrite(BuzzerPin,HIGH);
  pinMode (BuzzerPin,OUTPUT) ;
  tone(BuzzerPin,320,200);
}

void NoBee()
{
  digitalWrite(BuzzerPin, LOW);
  pinMode (BuzzerPin,INPUT) ;
  noTone(BuzzerPin);
}

void Fire_loop()
{
  boolean stat = digitalRead(DigitalInPin);
  if(stat == HIGH){
    NoBee();  //判断无事发生
    //Serial.println("Peace");
  }
  if(stat == LOW){
    Bee();  //判断有火灾
    //Serial.println("Fire");
  }
}
```

C. Photosensitive alarm

Besides fire, the theft of the warehouse also causes the loss of goods and poses a serious threat to the security of the warehouse. So, the monitoring system of the warehouse is essential.

Considering that thieves usually commit theft at night, we install a light sensor on our car, which can detect ambient lighting. If the light intensity is overly strong (possibly caused by the invader), the alarm will sound, in order to help warehouse managers, stop theft and reduce the loss of goods.

The code is shown below:

```
//定义火焰传感器、光敏模块与蜂鸣器引脚
#define DigitalInPin 26    //DO 引脚:  digital 26
#define BuzzerPin 27       //buzzer 引脚:  digital 27
#define idrpin    A5

//定义光敏模块数据范围
#define low=0;
#define high=1023;
```

```
void Guangmin_loop()
{
  int guangmin = analogRead(idrpin);   //读入光敏电阻产生的模拟信号（范围：0~1023）
  /*
  Serial.print("value= ");
  Serial.print(guangmin);
  */
  guangmin=map(guangmin,0,1023,255,0);  //map函数 将0~1023范围映射到255~0
  /*
  Serial.print("--");
  Serial.println(guangmin);
  */
  if(guangmin>=200)
    {
      Bee(); //判断有强光
    }
  else
    {
      NoBee();  //判断无事发生
    }
  delay(750);
}
```

D. Cargo identification

For warehouse management, patrolling and inspecting individual goods in the warehouse is the most basic work. Normally, warehouse managers can identify the kinds of goods by their different colors.

Based on this, we choose to use the TCS230 Color Sensor which can sense red, green, yellow and blue light to identify the color (category) of the goods. This color sensor can significantly reduce the workload of warehouse managers and automate warehouse cargo inspection.

It is worth mentioning that, because of the limitation of hardware, we optimize our algorithm which dramatically improves the identification accuracy. We combine our knowledge about HSV on a multimedia basis into coding, transforming RGB into HSV, finally set specific range for hardware to define the color.

The code is shown below:

```
void TSC_Init()
        {
        pinMode(S0Pin, OUTPUT);
        pinMode(S1Pin, OUTPUT);
        pinMode(S2Pin, OUTPUT);
        pinMode(S3Pin, OUTPUT);
        pinMode(OUTPin,INPUT);
        //OUTPUT Frequency Scaling 2%
        digitalWrite(S0Pin,LOW);
        digitalWrite(S1Pin,HIGH);
        }
//Select the filter color
void TSC_FilterColor(int Level01, int Level02)
{
  if(Level01 != 0)
  Level01= HIGH;
  if(Level02 != 0)
  Level02= HIGH;
  digitalWrite(S2Pin,Level01);
  digitalWrite(S3Pin,Level02);
}

//interrupt function, calculate TCS3200 output signal of pulse number
void TSC_Count(){g_count ++;}

void TSC_Callback()
{
  switch(g_flag)
  {
    case 0:
    Serial.println("->WB Start");
    TSC_WB(LOW, LOW);   //选择让红色光线通过滤波器的模式
    break;

    case 1:
    Serial.print("->Frequency R=");
    Serial.println(g_count);   //打印1s内的红光通过滤波器时，TCS3200输出的脉冲个数
    g_array[0] = g_count;    //存储1s内的红光通过滤波器时，TCS3200输出的脉冲个数
    TSC_WB(HIGH, HIGH);  //选择让绿色光线通过滤波器的模式
    break;

    case 2:
    Serial.print("->Frequency G=");
    Serial.println(g_count);   //打印1s内的绿光通过滤波器时，TCS3200输出的脉冲个数
    g_array[1] = g_count;    //存储1s内的绿光通过滤波器时，TCS3200输出的脉冲个数
    TSC_WB(LOW, HIGH);  //选择让蓝色光线通过滤波器的模式
    break;

    case 3:
    Serial.print("->Frequency B=");
    Serial.println(g_count);   //打印1s内的蓝光通过滤波器时，TCS3200输出的脉冲个数
    Serial.println("->WB End");
    g_array[2] = g_count;     //存储1s内的蓝光通过滤波器时，TCS3200输出的脉冲个数
    TSC_WB(HIGH, LOW);   //选择无滤波器的模式
    break;

    default:
    g_count = 0;     //计数值清零
    break;
  }
}
//设置反射光中红、绿、蓝三色光分别通过滤波器时如何处理数据的标志

//该函数被TSC_Callback( )调用
void TSC_WB(int Level0, int Level1)
{
  g_count = 0;   //计数值清零
  g_flag ++;     //输出信号计数标志
  TSC_FilterColor(Level0, Level1); //滤波器模式
  Timer1.setPeriod(1000000);     //设置输出信号脉冲计数时长1s
}

void Color_set()
{
  TSC_Init();
```

```
  Timer1.initialize();   // defaulte is 1s
  Timer1.attachInterrupt(TSC_Callback); //设置定时器1的中断，中断调用函数为TSC_Callback()
  //设置TCS3200输出信号的上跳沿触发中断，中断调用函数为TSC_Count()
  attachInterrupt(0, TSC_Count, RISING);
  delay(4000); //延时4s，以等待被测物体红、绿、蓝三色在1s内的TCS3200输出信号脉冲计数
  //通过白平衡测试，计算得到白色物体RGB值255与1s内三色光脉冲数的RGB比例因子
  g_SF[0] = 255.0/ g_array[0];     //红色光比例因子
  g_SF[1] = 255.0/ g_array[1] ;    //绿色光比例因子
  g_SF[2] = 255.0/ g_array[2] ;    //蓝色光比例因子
  //打印白平衡后的红、绿、蓝三色的RGB比例因子
  Serial.println(g_SF[0],5);
  Serial.println(g_SF[1],5);
  Serial.println(g_SF[2],5);//红、绿、蓝三色光分别对应的1s内TCS3200输出脉冲数乘以相应的比例因子就是RGB标准值。打印被测物体的RGB值

  for(int i=0; i<3; i++)
    Serial.println(int(g_array[i] * g_SF[i]));
}

void Color_loop()
{
  g_flag = 0;
  //每获得一次被测物体RGB颜色值需时4s
  delay(4000);
  //打印出被测物体RGB颜色值
  for(int i=0; i<3; i++){
    Serial.println(int(g_array[i] * g_SF[i]));
    compare[i]=int(g_array[i] * g_SF[i]);
  }

  if(compare[0]>compare[1])
  {
    numMax=compare[0];
  }
  else
  {
    numMax=compare[1];

  }
  if(compare[2]>numMax)
  {
    numMax = compare[2];
  }

  if(compare[0]<compare[1])
  {
    numMin=compare[0];
  }
  else
  {
    numMin=compare[1];
  }
  if(compare[2]<numMin)
  {
    numMin=compare[2];
  }

  if(numMin<=110&&numMax<70)  //小车自动认定为未检测到货物，并且不进行后续颜色判断
  {
    Serial.println("Don't have any object being detected");//修改none判定
    lcd.setCursor(8,0);
    lcd.print("None   ");
  }
  else
  {
    delta=numMax-numMin;
    if(numMax==compare[0])
    {
      H=(compare[1]-compare[2])*60/delta;
    }
    else if(numMax==compare[1])
    {
      H = 120 + (compare[2]-compare[0])*60/delta;
    }
    else
    {
      H = 240 + (compare[0]-compare[1])*60/delta;
```

```
    }

    if(H<0)
      H+=360;

    if(H<30||H>300)
    {
      Serial.println("Red");  //检测为红色
      lcd.setCursor(8,0);
      lcd.print("Red    ");
    }
    else if(H>=30 && H<=80)
    {
      Serial.println("Yellow");  //检测为黄色
      lcd.setCursor(8,0);
      lcd.print("Yellow ");
    }
    else if(H>80 && H<=180)
    {
      Serial.println("Green");  //检测为绿色
      lcd.setCursor(8,0);
      lcd.print("Green  ");
    }
    else if(H>=180 && H<=280)
    {
      Serial.println("Blue");  //检测为蓝色
      lcd.setCursor(8,0);
      lcd.print("Blue   ");
    }

    Serial.println(H);
  }
}
```

E. Infrared tracking

Patrolling warehouses is one of the basic jobs of a warehouse manager. And to automate warehouse inspections, we decide to let our car follow the given path (the black line on the ground) to patrol the warehouse.

To enable the car to sensor the black line around and be able to follow it, an infrared tracking sensor is installed on the car, which helps the car adjust the servo speed according to the recognized path, in order to follow the given road.

The code is shown below:

```
void xunji_loop()
{
    SL = digitalRead(SensorLeft);
    SM = digitalRead(SensorMiddle);
    SR = digitalRead(SensorRight);
        Serial.print(SL);
        Serial.print("-");
        Serial.print(SM);
        Serial.print("-");
        Serial.println(SR);
        if (SM == HIGH)//中间传感器检测到黑线  黑线为1 白为0
    {
        if (SL == LOW & SR ==  HIGH) //左白右黑, 向右转弯
        {
            Serial.println("BL");
            analogWrite(MotorRight1, 0); //右转
            analogWrite(MotorRight2, 80);
            digitalWrite(MotorLeft1, LOW);
            digitalWrite(MotorLeft2, HIGH);
        }
        else if ( SL ==  HIGH & SR == LOW) // 左黑右白, 向左转弯
        {
            Serial.println("BR");
```

```
            digitalWrite(MotorRight1, LOW);
            digitalWrite(MotorRight2, HIGH);
            analogWrite(MotorLeft1, 0);
            analogWrite(MotorLeft2, 80);
        }
        else if ( SL == LOW & SR == LOW)  // 两侧为白色，前进
        {
            Serial.println("Bw");
            digitalWrite(MotorRight1, LOW);
            digitalWrite(MotorRight2, HIGH);
            digitalWrite(MotorLeft1, LOW);
            digitalWrite(MotorLeft2, HIGH);
            /*
              analogWrite(MotorLeft1,0);
              analogWrite(MotorLeft2,150);
              analogWrite(MotorRight1,0);
              analogWrite(MotorRight2,150);
            */
        }
        else if ( SL == HIGH & SR == HIGH)
        {
            Serial.println("Bb");
            advance(1);
            stopp(20);


            digitalWrite(MotorRight1, LOW);
            digitalWrite(MotorRight2, HIGH);
            digitalWrite(MotorLeft1, LOW);
            digitalWrite(MotorLeft2, HIGH);

        }
    }
        else // 中间传感器在白色区域
    {
        if (SL == LOW & SR == HIGH ) // 左白右黑，快速右转
        {
            Serial.println("WL");

            digitalWrite(MotorRight1, LOW);
            digitalWrite(MotorRight2, LOW);
            digitalWrite(MotorLeft1, LOW);
            digitalWrite(MotorLeft2, HIGH);
        }
        else if ( SL == HIGH & SR == LOW ) // 左黑右白，快速左转
        {
            Serial.println("WR");

            digitalWrite(MotorRight1, LOW);
            digitalWrite(MotorRight2, HIGH);
            digitalWrite(MotorLeft1, LOW);
            digitalWrite(MotorLeft2, LOW);
        }
        else // 都是白色，保持不变
        {
            Serial.println("ww");
            /*
                        analogWrite(MotorLeft1,255);
                        analogWrite(MotorLeft2,125);
                        analogWrite(MotorRight1,255);
                        analogWrite(MotorRight2,125);
            */
        }
    }
}
```

F. Infrared obstacle avoidance

In the process of automatically patrolling the warehouse along the path, sometimes there might be some obstacles on the road. Hence, our car should be able to detect obstacles ahead in real-time, control the motor to slow down, then stop. To realize the functions mentioned above, we use Infrared obstacle avoidance, which can ensure the obstacle avoidance ability of the car during the automatic inspection, as well as improve the flexibility and stability of patrols.

The code is shown below:

```
void hongwaibizhang_loop()
{
  if ((digitalRead(barrier_left)==LOW) && (digitalRead(barrier_right)==LOW))
        {
          back(5);//后退
        }
        else if (digitalRead(barrier_left)==LOW)
        {
          right(6); // 右转;
        }
        else if (digitalRead(barrier_right)==LOW)
        {
          left(6); // 左转;
        }
        else
          advance(0);//前进
}
```

G. Ultrasonic ranging and warehouse counting

Whether obstacle avoidance or cargo identification, it is necessary for the car to detect the distance between objects (goods or balks) and count the number of the objects. So we choose HC-SR04, which can emit ultrasonic waves and receive ultrasonic waves reflected back from the object, judging the distance between the trolley and the object according to the time difference. In addition, according to the distance detected, some functions like obstacle avoidance or cargo identification can be triggered automatically.

The code is shown below:

```
//设置LCD液晶屏幕以及超声波探测模块引脚
LiquidCrystal_I2C lcd(0x27,16,2); //设置LCD1_602A设备地址
UltraSonicDistanceSensor distanceSensor(24, 25);  // 初始化引脚，第一个参数为Trig引脚，第二个参数为Echo引脚。
int block_count = 0;

void Ultrasound()
{
  //Serial.println(distanceSensor.measureDistanceCm());//打印超声波测量距离，单位为厘米
  lcd.setCursor(0,0);            //第一行显示
  lcd.print(distanceSensor.measureDistanceCm());       //输出字符
  lcd.setCursor(5,0);
  lcd.print("cm");
  if(distanceSensor.measureDistanceCm() <= 10)
  {
    block_count++;  //距离判断后根据位置计数 仓库数会随着行进累加
    lcd.setCursor(15,0);
    lcd.print(block_count);
    if(block_count>=4){ //当仓库数大于4的时候，默认循迹完成，仓库数清零
      block_count=0;
    }
  }
}
```

H. LCD visualization

To show the comprehensive information about the warehouse to the manager of the warehouse, we display the temperature and humidity detected by the car, the distance from the object, and the color detection results on the LCD screen of the car, which is convenient for the supervisor to gather such information.

The code is shown below:

```
//设置LCD液晶屏幕以及超声波探测模块引脚
LiquidCrystal_I2C lcd(0x27,16,2); //设置LCD1_602A设备地址

void setup()
{
  Serial.begin(9600);                          // 打开串口监视器
  Color_set();
  lcd.init();                                 // 初始化LCD_1602A
  lcd.backlight();                            //设置LCD背景等亮
  dht.begin();                                 //DHT开始工作

  Bee_set();
}

void Ultrasound()
{
  //Serial.println(distanceSensor.measureDistanceCm());//打印超声波测量距离，单位为厘米
  lcd.setCursor(0,0);             //第一行显示
  lcd.print(distanceSensor.measureDistanceCm());       //输出字符
  lcd.setCursor(5,0);
  lcd.print("cm");
  if(distanceSensor.measureDistanceCm() <= 10)
  {
    block_count++;
    lcd.setCursor(15,0);
    lcd.print(block_count);
    if(block_count>=4){
      block_count=0;
    }
  }
}

void TandH()
{
  lcd.setCursor(0,1);             //第二行显示
  lcd.print(dht.readHumidity());       //输出字符
  lcd.setCursor(5,1);
  lcd.print("%");

  lcd.setCursor(7,1);
  lcd.print(dht.readTemperature());       //输出字符
  lcd.setCursor(12,1);
  lcd.print(char(0xdf));
  lcd.setCursor(13,1);
  lcd.print("C");
}
```

I. Wi-Fi connection upload

This part is the most significant one. For further information gathering and warehouse managing, we have designed the Wi-Fi transmission module to upload information like the temperature, humidity, the color of the goods, the distance to the object, and abnormal conditions (fire, light) to the database of cloud platform by the. json format file. Warehouse managers can view such information on the software.

The code is shown below:

```
#include <Arduino.h>
#include "WiFi.h"
#include "PubSubClient.h"
#include "Ticker.h"
//#include "uFire_SHT20.h"
#include "ArduinoJson.h"

//uFire_SHT20 sht20;

const char* ssid = "HUAWEI P40";              //wifi名
const char* password = "babababa";      //wifi密码
const char* mqtt_server = "mqtts.heclouds.com"; //onenet 的 IP地址 mqtts.heclouds.com 183.230.40.96
const int port = 1883;                    //端口号

#define mqtt_pubid "550563"    //产品ID
#define mqtt_devid "hby_1"       //设备名称
//鉴权信息
#define mqtt_password "version=2018-10-31&res=products%2F550563%2Fdevices%2Fhby_1&et=1697699055&method=md5&sign=eb4cZ%2BiZqa%2BLz8RLLBDIQQ%3D%3D"       //鉴权信息

WiFiClient espClient;           //创建一个WIFI连接客户端
PubSubClient client(espClient); // 创建一个PubSub客户端, 传入创建的WIFI客户端
Ticker tim1;                    //定时器,用来循环上传数据
Ticker tim2;                    //定时器,用来循环上传数据
float temp;
float humi;
float dis;


//设备上传数据的post主题
#define ONENET_TOPIC_PROP_POST "$sys/" mqtt_pubid "/" mqtt_devid "/dp/post/json" //"$sys/" mqtt_pubid "/" mqtt_devid "/thing/property/post"
//接收下发属性设置主题
#define ONENET_TOPIC_PROP_SET  "$sys/" mqtt_pubid "/" mqtt_devid "/dp/post/json/+" //"$sys/" mqtt_pubid "/" mqtt_devid "/thing/property/set"
//接收下发属性设置成功的回复主题
#define ONENET_TOPIC_PROP_SET_REPLY "$sys/" mqtt_pubid "/" mqtt_devid "/dp/post/accepted" //"$sys/" mqtt_pubid "/" mqtt_devid "/thing/property/set_reply"

//接收设备属性获取命令主题
#define ONENET_TOPIC_PROP_GET "$sys/" mqtt_pubid "/" mqtt_devid "/cmd/request/+" //"$sys/" mqtt_pubid "/" mqtt_devid "/thing/property/get"
//接收设备属性获取命令成功的回复主题
#define ONENET_TOPIC_PROP_GET_REPLY "$sys/" mqtt_pubid "/" mqtt_devid "/cmd/response/+/+" //"$sys/" mqtt_pubid "/" mqtt_devid "/thing/property/get_reply"

//这是post上传数据使用的模板
#define ONENET_POST_BODY_FORMAT "{\"id\":%d,\"dp\":%s}"
//#define ONENET_POST_BODY_FORMAT
int postMsgId = 0; //记录已经post了多少条
```

```
//连接WIFI相关函数
void setupWifi()
{
  delay(10);
  Serial.println("connect WIFI");
  WiFi.begin(ssid, password);
  while (!WiFi.isConnected())
  {
    Serial.print(".");
    delay(500);
  }
  Serial.println("OK");
  Serial.println("Wifi connected!");
  Serial.println("IP address: ");
  Serial.println(WiFi.localIP());

}

//重连函数, 如果客户端断线,可以通过此函数重连
void clientReconnect()
{
  while (!client.connected()) //再重连客户端
  {
    Serial.println("reconnect MQTT...");
    if ( client.connect(mqtt_devid, mqtt_pubid, mqtt_password) )
    {
      Serial.println("connected");
    }
    else
    {
      Serial.println("failed");
      Serial.println(client.state());
      Serial.println("try again in 5 sec");
      delay(5000);
    }
  }
}

void callback(char *topic, byte *payload, unsigned int length)
{
  Serial.println("message rev:");
  Serial.println(topic);
  for (size_t i = 0; i < length; i++)
  {
    Serial.print((char)payload[i]);
  }
  Serial.println();
```

```
    if ( client.connect(mqtt_devid, mqtt_pubid, mqtt_password) )
    {
      Serial.println("connected");
    }
    else
    {
      Serial.println("failed");
      Serial.println(client.state());
      Serial.println("try again in 5 sec");
      delay(5000);
    }
  }
}

void callback(char *topic, byte *payload, unsigned int length)
{
  Serial.println("message rev:");
  Serial.println(topic);
  for (size_t i = 0; i < length; i++)
  {
    Serial.print((char)payload[i]);
  }
  Serial.println();


  if (strstr(topic, ONENET_TOPIC_PROP_SET))
  {
    DynamicJsonDocument doc(100);
    DeserializationError error = deserializeJson(doc, payload);
    if (error)
    {
      Serial.println("parse json failed");
      return;
    }
    JsonObject setAlinkMsgObj = doc.as<JsonObject>();
    serializeJsonPretty(setAlinkMsgObj, Serial);
    String str = setAlinkMsgObj["id"];
    Serial.println(str);
    char sendbuf[100];
    sprintf(sendbuf, "{\"id\": \"%s\",\"code\":200,\"msg\":\"success\"}", str.c_str());
    Serial.println(sendbuf);
    client.publish(ONENET_TOPIC_PROP_SET_REPLY, sendbuf);
  }
  if (strstr(topic, ONENET_TOPIC_PROP_GET))
  {
    DynamicJsonDocument doc(100);
    DeserializationError error = deserializeJson(doc, payload);
    if (error)
    {
      Serial.println("parse json failed");
      return;
    }

    JsonObject setAlinkMsgObj = doc.as<JsonObject>();
    serializeJsonPretty(setAlinkMsgObj, Serial);
    String str = setAlinkMsgObj["id"];
    Serial.println(str);
    char sendbuf[100];
    //sprintf(sendbuf, "{\"id\": \"%s\",\"code\":200,\"msg\":\"success\",\"data\":{\"temp\":%.2f,\"humi\":%.2f}}", str.c_str(), temp, humi);
    sprintf(sendbuf, "{\"id\": \"%s\",\"code\":200,\"msg\":\"success\",\"data\":{\"dis\":%.2f }}", str.c_str(), temp);
    Serial.println(sendbuf);
    client.publish(ONENET_TOPIC_PROP_GET_REPLY, sendbuf);
  }
}

void setup() {
  // put your setup code here, to run once:


  Serial.begin(115200); //初始化串口
  Serial2.begin(9600);
  //Wire.begin();
  //sht20.begin();
  delay(3000);
  setupWifi();                                        //调用函数连接WIFI
  client.setServer(mqtt_server, port);                //设置客户端连接的服务器,连接Onenet服务器, 使用1883端口
  delay(2000);
  Serial.println("setServer Init!");
  //client.setCallback(callback);
  client.connect(mqtt_devid, mqtt_pubid, mqtt_password); //客户端连接到指定的产品的指定设备.同时输入鉴权信息
  delay(2000);
  Serial.println("connect Init!");

  if (client.connected())
  {
    Serial.println("OneNet is connected!"); //判断以下是不是连好了.
  }
  client.subscribe(ONENET_TOPIC_PROP_SET);
  client.subscribe(ONENET_TOPIC_PROP_GET);
  client.setCallback(callback);
  tim1.attach(8, sendTempAndHumi); //定时每20秒调用一次发送数据函数sendTempAndHumi
  tim2.attach(4, update_date);
```

```
  temp =10;
  humi =20;
  dis = 100;
}
int count = 0;

void loop() {
  // put your main code here, to run repeatedly:

  if(Serial2.available()>0){
    char tmp = Serial2.read();
    Serial2.print(tmp);
    Serial.println("recieved");
  }
  if (!WiFi.isConnected()) //先看WIFI是否还在连接
  {
    setupWifi();
  }
  if (!client.connected()) //如果客户端没连接ONENET，重新连接
  {
    clientReconnect();
    delay(100);
  }
  client.loop(); //客户端循环检测

}
```

J. Other innovative points

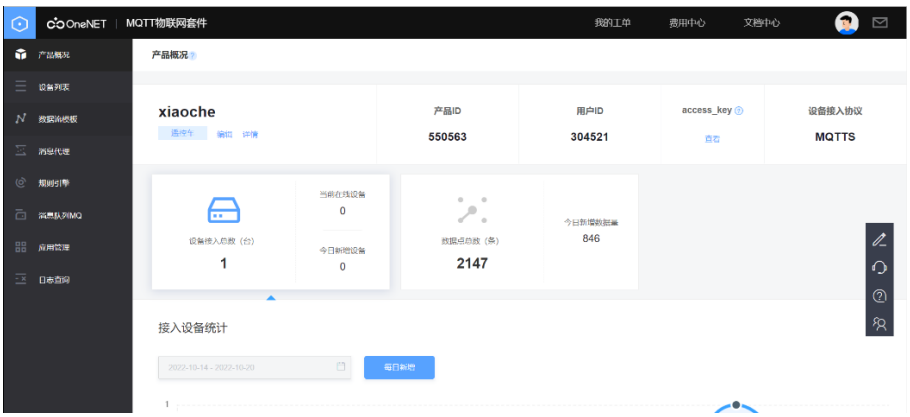
- **Data transfer platform – OneNET**

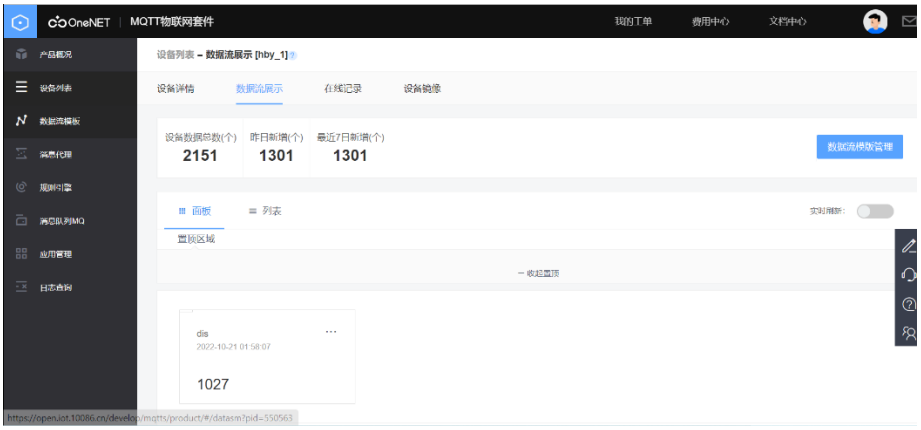
OneNET is dedicated to building an efficient, stable, and secure application platform between IoT applications and real devices. The platform adapts to a variety of network environments and common transmission protocols, providing fast access solutions and device management services for various hardware terminals. It also provides rich APIs and data distribution capabilities to meet the development needs of various industry applications, enabling IoT enterprises to focus more on their own application development.



MQTT IoT Suite is a function of OneNET, which provides developers with effective, creative, and safe device access services. While meeting the needs of multiple application scenarios, it ensures reliable data transmission.

We choose the MQTT IoT Suite service as a transit point for data collection and writing to the database. We guarantee stable data transmission while using a more flexible Wi-Fi communication module, reducing the difficulty of interfacing Electronic Engineering and Management with IoT and E-commerce & Law.







(5) Summary

All in all, this car has some advantages like a good-looking appearance, multiple comprehensive functions, automatically supervising the warehouse and responding to the manager on time, good data collection and updating, lower cost as well as better commercial use, etc. For the manager of the warehouse, this car just plays the role of a good assistant, like a jack of all trades.

2.2.2. Software

- **Environment**

Front End

```
"dependencies": {
    "axios": "^0.21.1",
    "echarts": "^5.3.3",
    "element-plus": "^1.0.2-beta.52",
    "vue": "^3.2.4",
    "vue-cropperjs": "^5.0.0",
    "vue-router": "^4.0.11",
    "vue-schart": "^2.0.0"
},
"devDependencies": {
    "@vue/cli-plugin-babel": "~4.5.0",
    "@vue/cli-plugin-router": "~4.5.0",
    "@vue/cli-plugin-vuex": "~4.5.0",
    "@vue/cli-service": "~4.5.0",
    "@vue/compiler-sfc": "^3.2.4"
},
```

Back End

```
<dependencies>
    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-web</artifactId>
    </dependency>
    <dependency>
        <groupId>org.mybatis.spring.boot</groupId>
        <artifactId>mybatis-spring-boot-starter</artifactId>
        <version>2.2.2</version>
    </dependency>
    <dependency>
        <groupId>mysql</groupId>
        <artifactId>mysql-connector-java</artifactId>
        <scope>runtime</scope>
    </dependency>
    <dependency>
        <groupId>org.projectlombok</groupId>
        <artifactId>lombok</artifactId>
        <optional>true</optional>
    </dependency>
    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-test</artifactId>
        <scope>test</scope>
```

```
        </dependency>
        <dependency>
            <groupId>com.baomidou</groupId>
            <artifactId>mybatis-plus-boot-starter</artifactId>
            <version>3.4.3.1</version>
        </dependency>
        <dependency>
            <groupId>cn.hutool</groupId>
            <artifactId>hutool-all</artifactId>
            <version>5.8.5</version>
        </dependency>
        <dependency>
            <groupId>com.auth0</groupId>
            <artifactId>java-jwt</artifactId>
            <version>3.10.3</version>
        </dependency>
</dependencies>
```

- **Website Function**

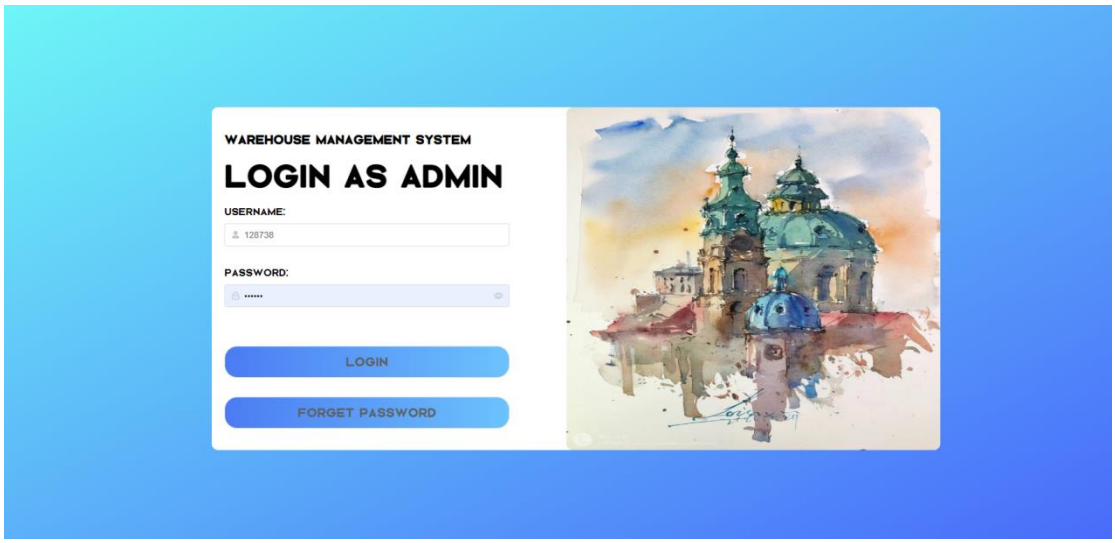
(1) Home Page

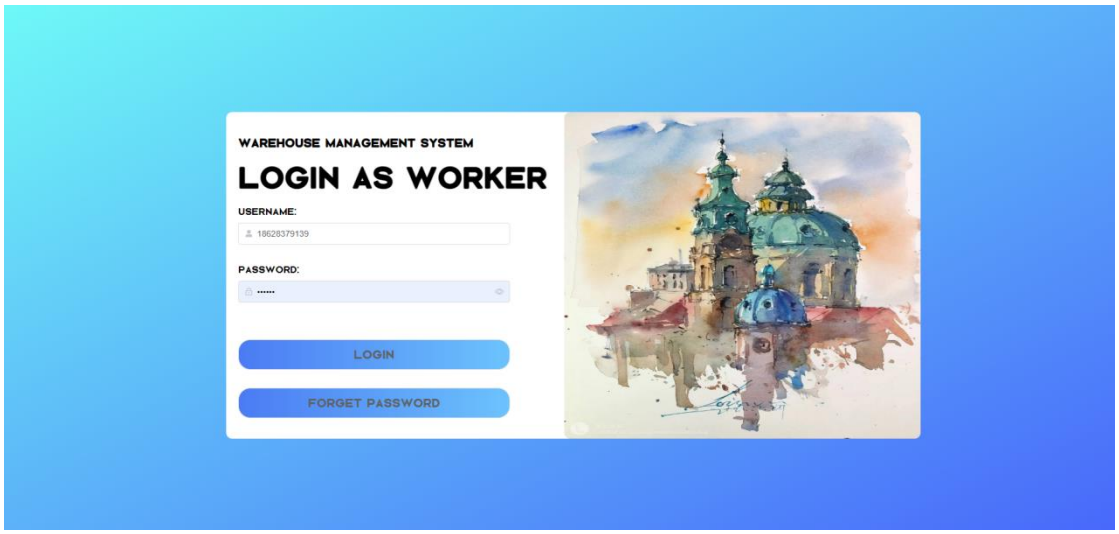
Login as the worker or the administrator.



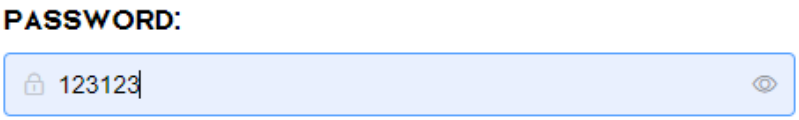
(2) Login

Enter the correct username and password to log in.





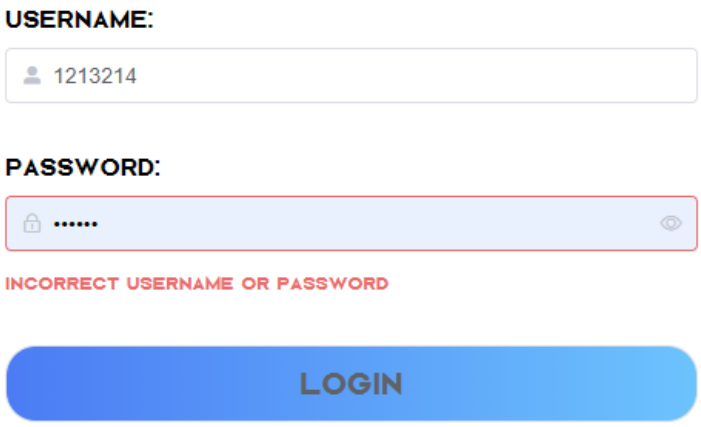
The password is hidden by default. But you can click the little "eye" button on the right to show your password.



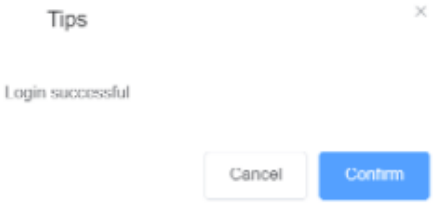
The length of the username should be between 5 and 10.



If the username and the password are not matched, you cannot log in.

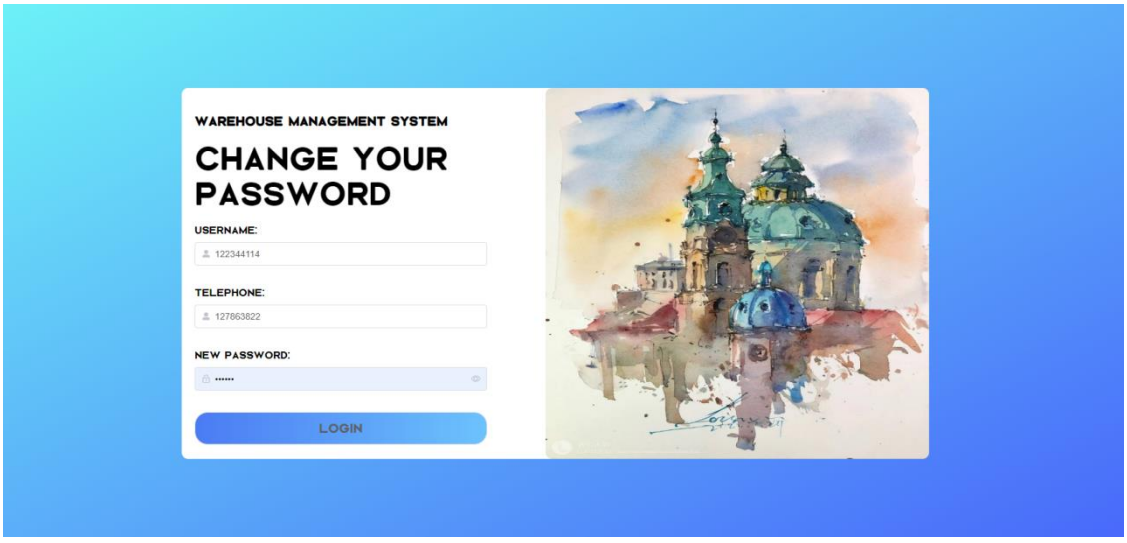


If you login successfully, the website will prompt you that you have successfully logged in.



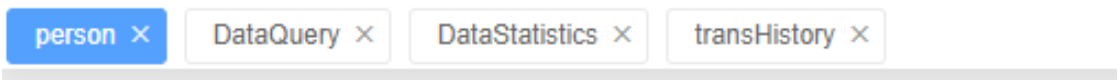
(3) Change Password

Users can change their passwords by providing the correct telephone numbers.

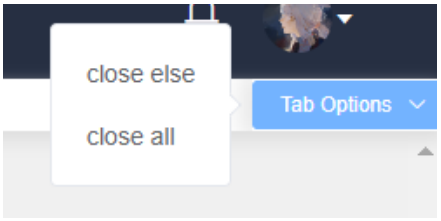


(4) Basic Interface Functions

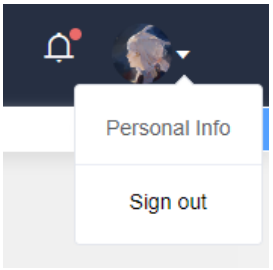
We provide tags to switch between different pages.



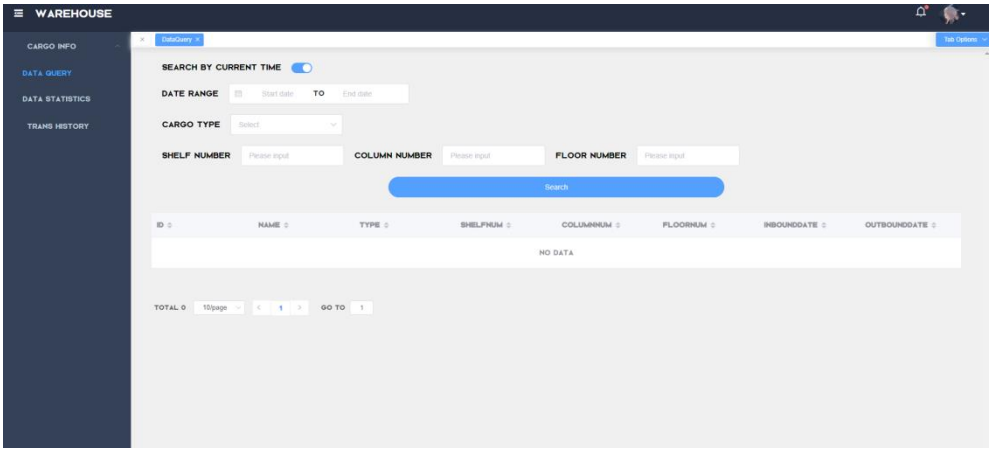
Also, there are "close all" and "close else" functions on the right of the navigate bar.



You can sign out by clicking the avatar.



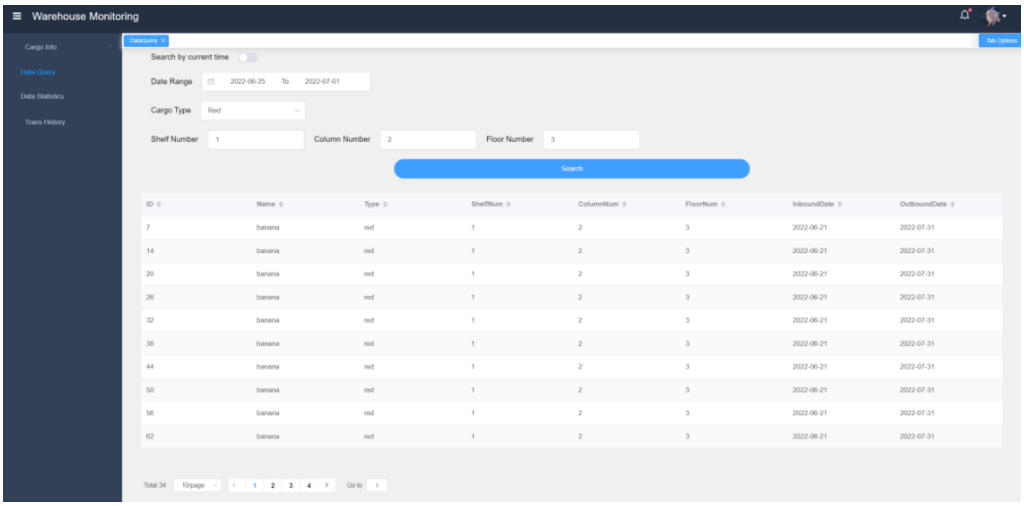
(5) Worker Interface Data Query



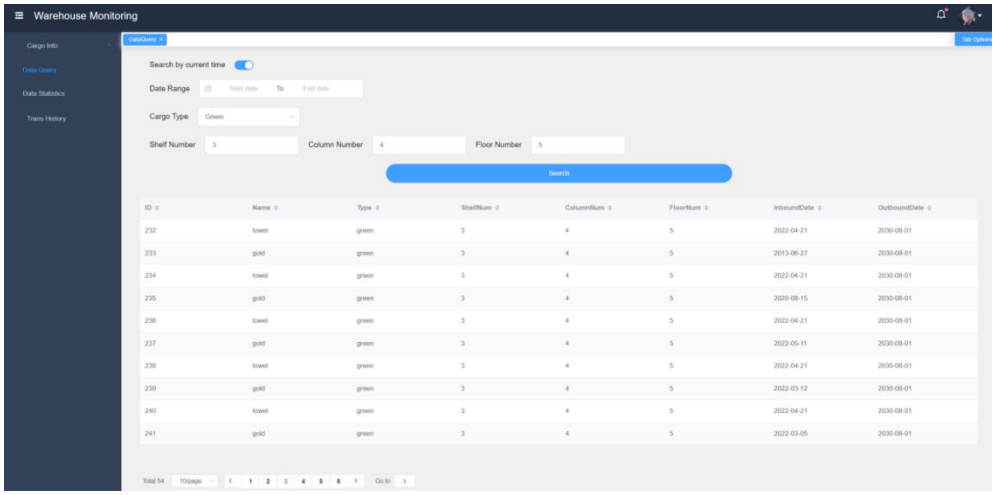
You can query by current date/any time, type of cargo, and location (shelf, column, floor). The query result can be paged, the number of data displayed on each page can be changed, and the page number can be mouse clicked or the specified number of pages can be entered directly to a page. Each column in the query result can be sorted (ascending/descending order).

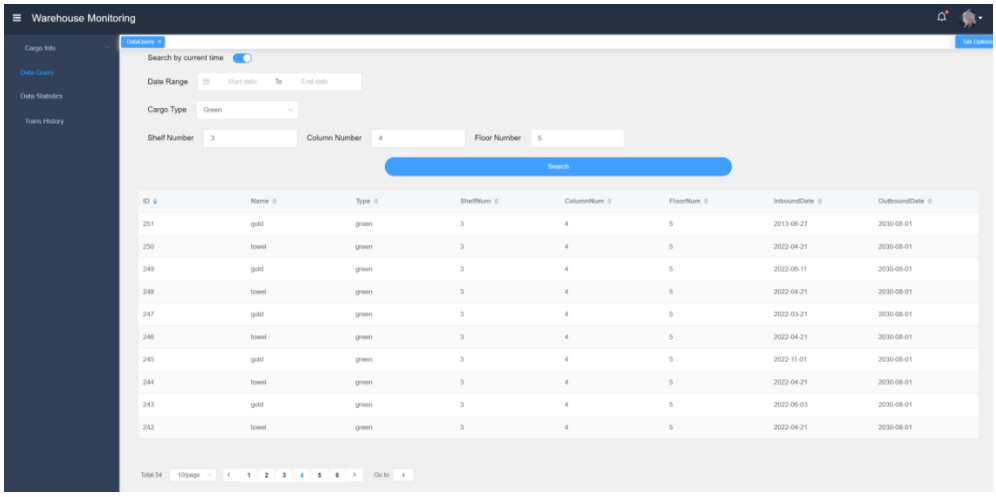
(6) Worker Interface Cargo Information

When we search for type = red, shelfNum = 1, ColumnNum = 2, FloorNum = 3 during 2022-06-25 ~ 2022-07-01, the webpage can display the following information:



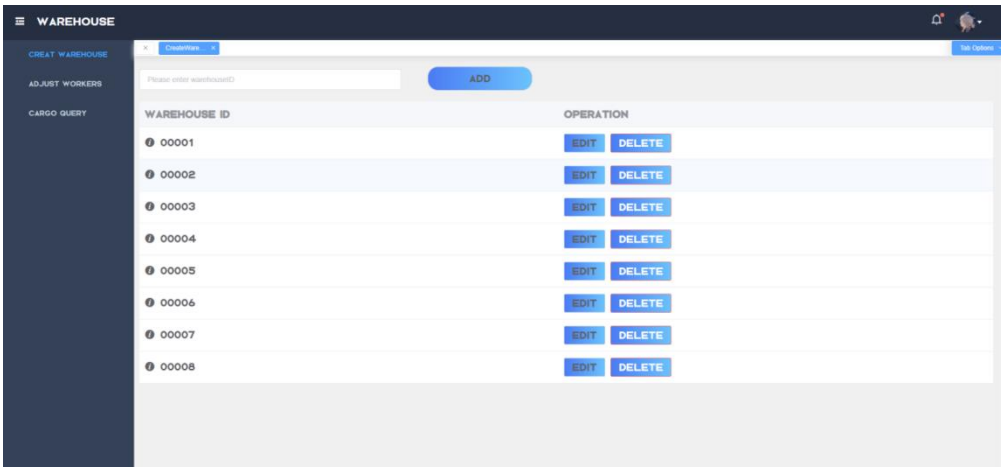
When we search all cargoes with type = green, shelfNum = 3, ColumnNum = 4, FloorNum = 5 by current time, the information will update to:





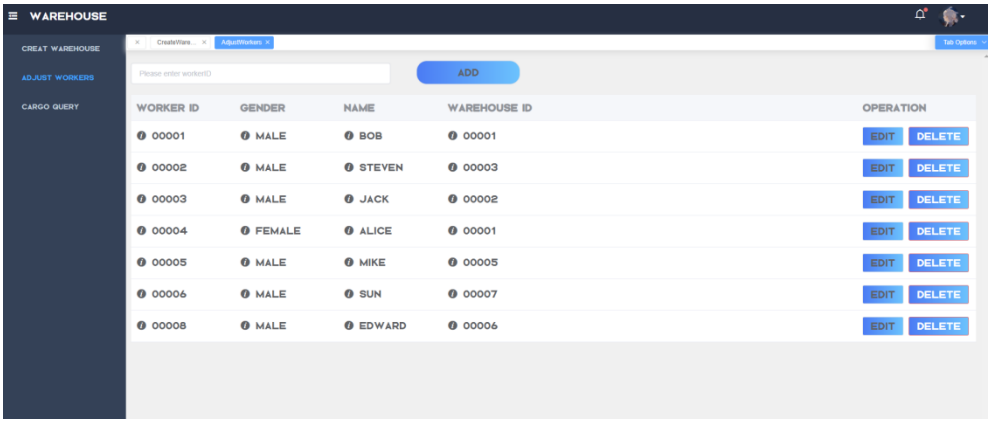
(7) Administrator Interface Create Warehouse

Administrator can edit and create workhouse by entering its ID.

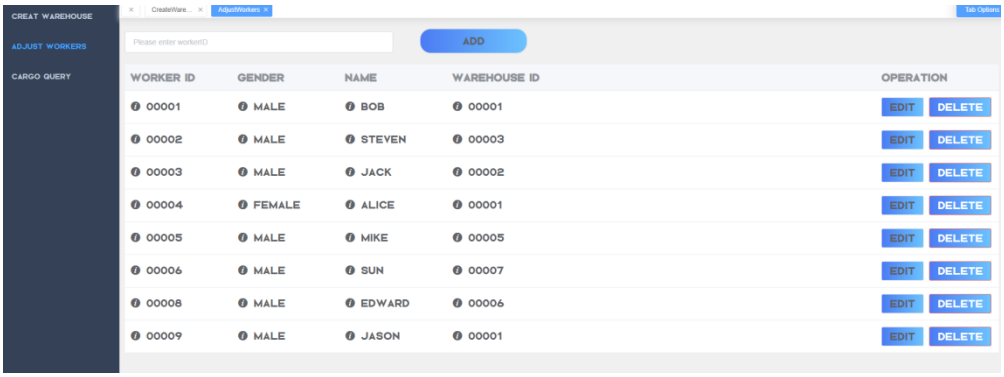


(8) Administrator Interface Adjust Workers

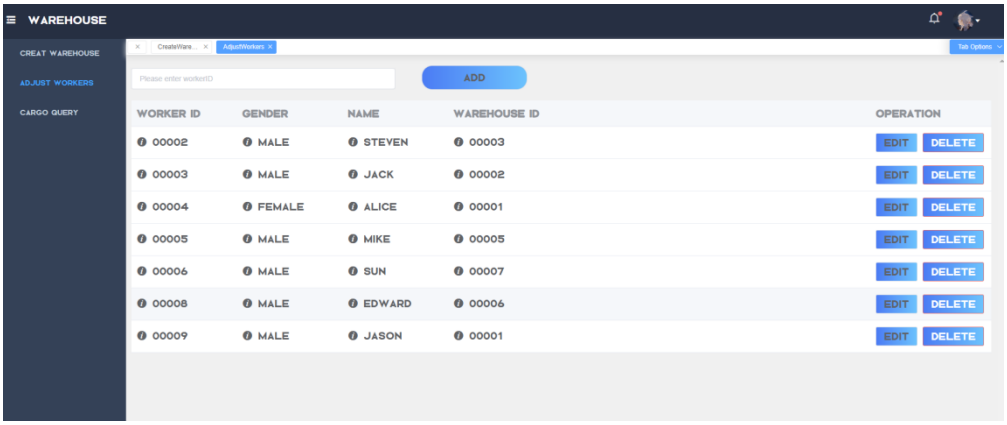
Administrator can add or delete workers, as well as edit their basic information.



If we want to add a new worker with values (00009, Male, Jason, 00001):

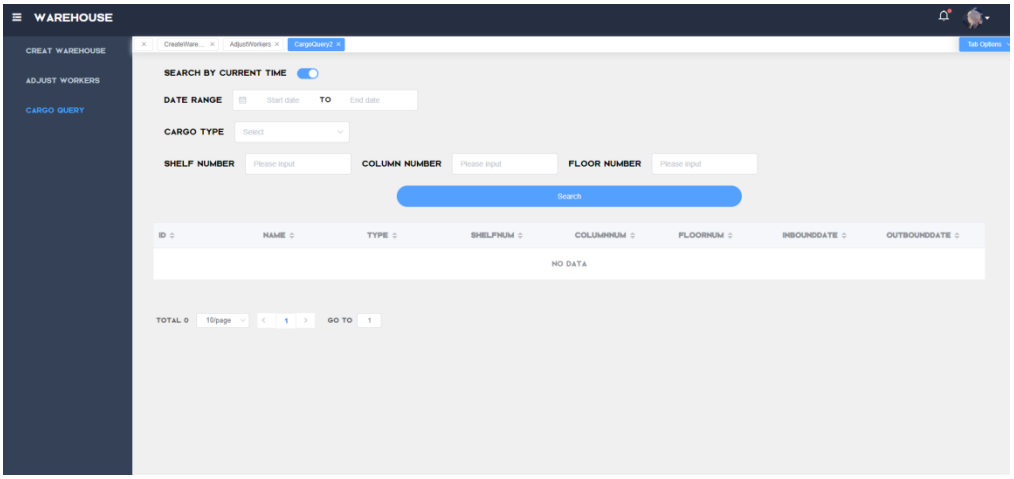


If we want to delete the worker whose ID is 00001:



(9) Admin Interface Cargo Query

Same as workers, administrator can query the information of cargoes.

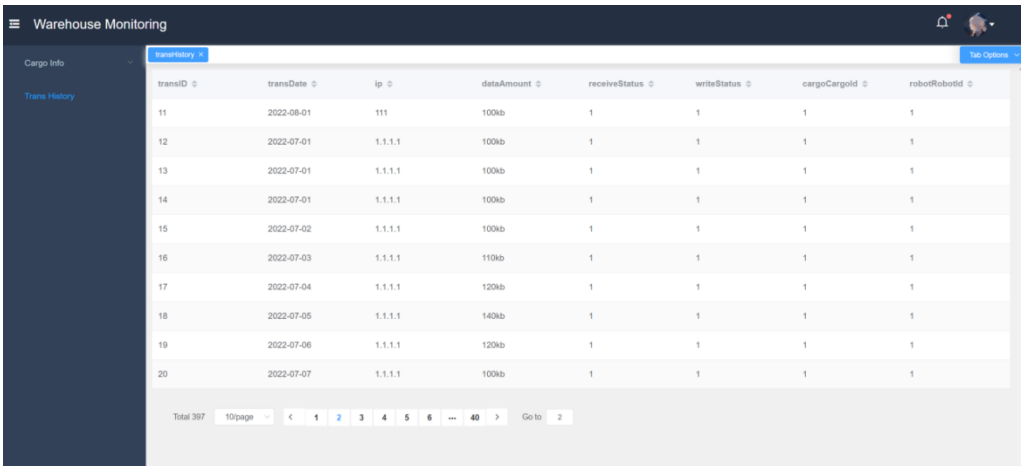


(10) Transmission History

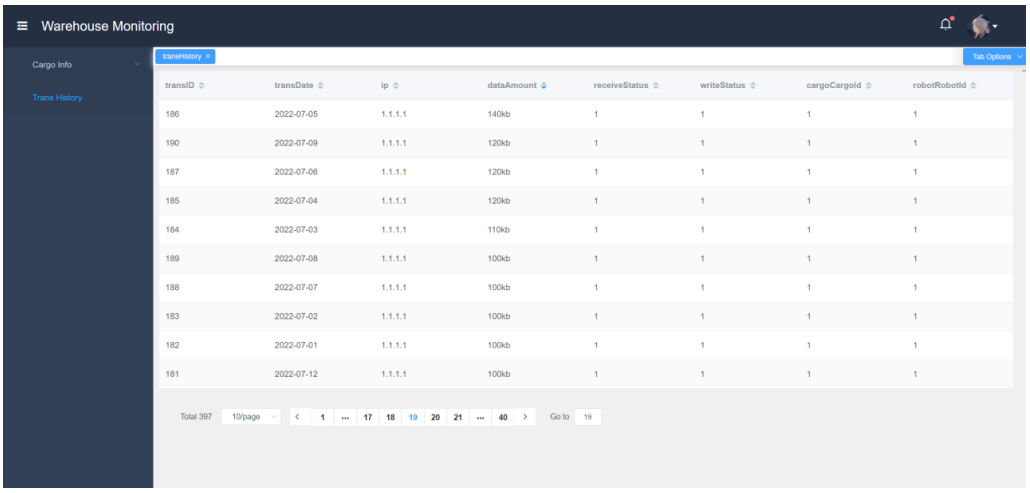
If we want to display all transmission history, we can turn the page through changing the page number, which can be done just by mouse clicking or typing the specified page number to reach a certain page.

The query results can be sorted in each column (ascending/descending).

From the following pictures, 10 items are shown on one page (default).

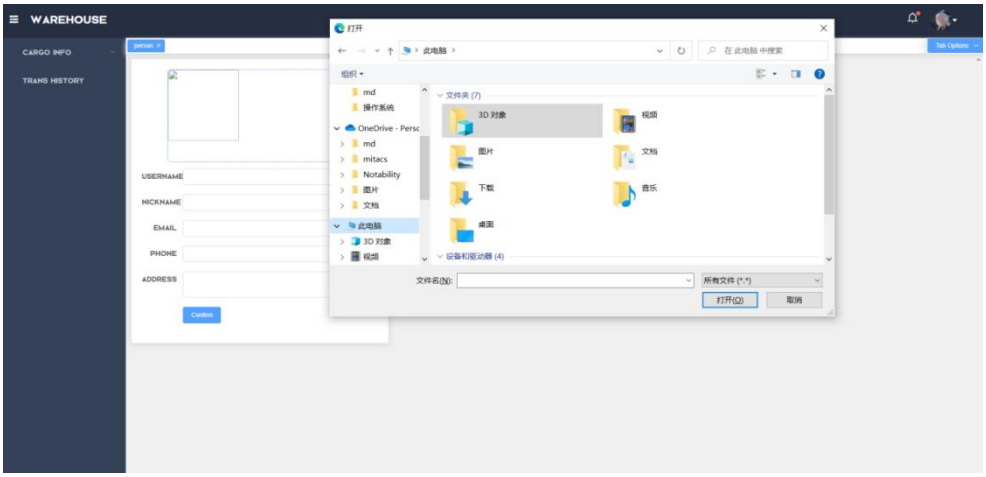


Then when the statistics are sorted by (dataAmount):



(11) Personal Information

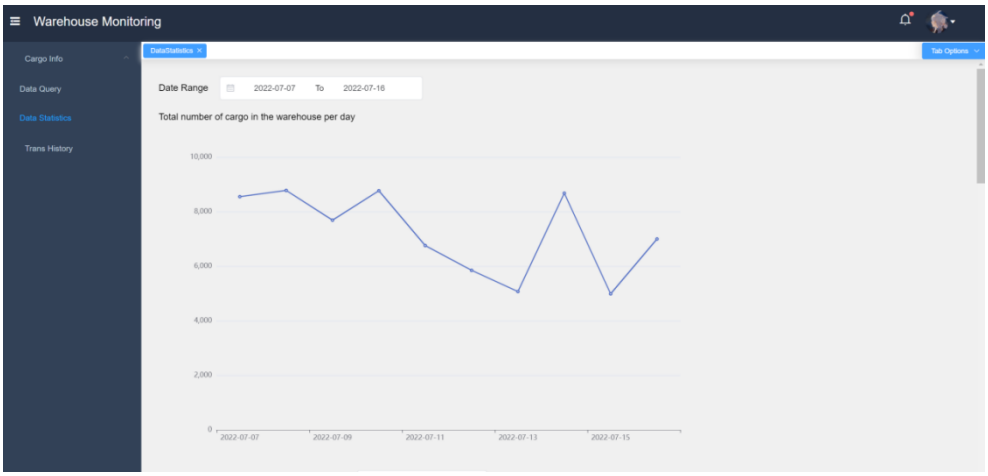
The worker can also change his username, nickname, email, phone, address, and upload the avatar. It's really personalized and aimed at satisfying all users.



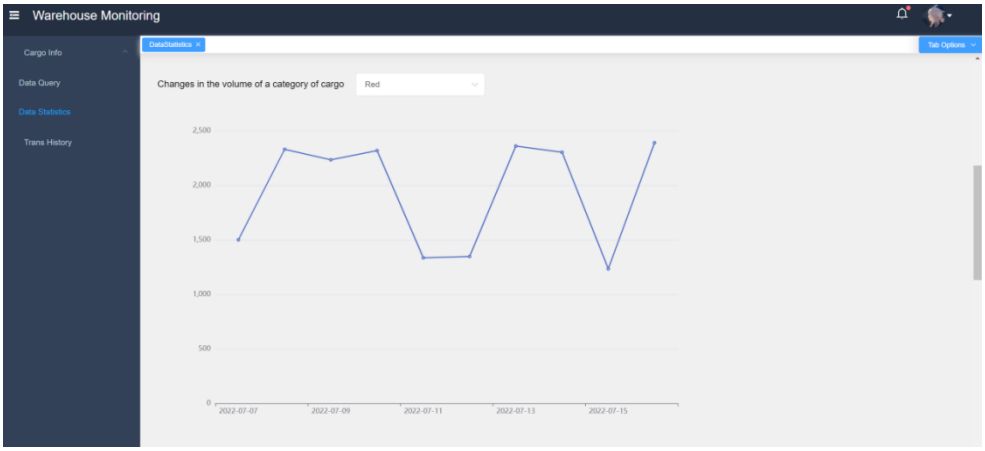
(12) Data Visualization

Workers can get a line chart of the change in the total number of cargoes in the warehouse every day for a specified period, a line chart shown the change in the number of cargoes of a certain category, a pie chart about the proportion of cargoes in different categories (taking the average of the period), and a line chart which displays the shift in the number of cargoes at different locations.

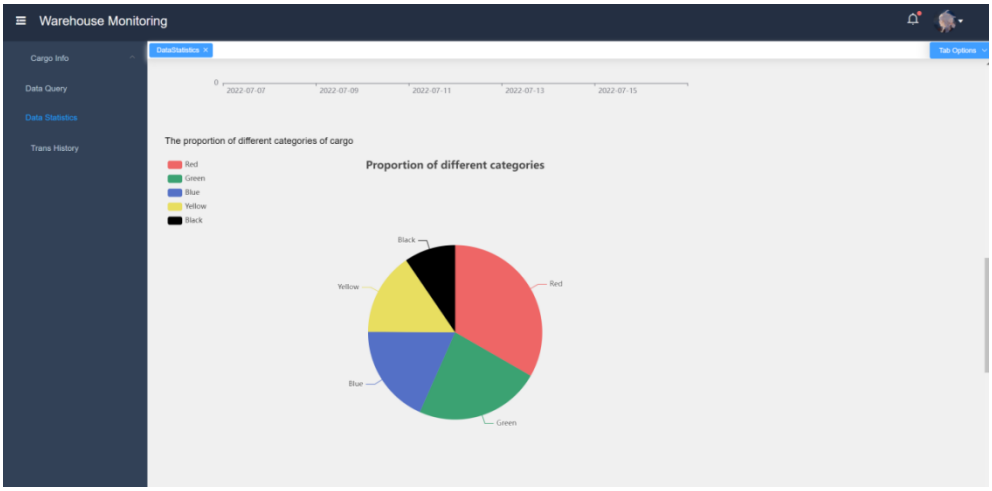
The following picture is a line graph, which visualize the shift about the total quantity of goods in the warehouse clearly when we select date range: 2022-07-07 ~ 2022-07-16.



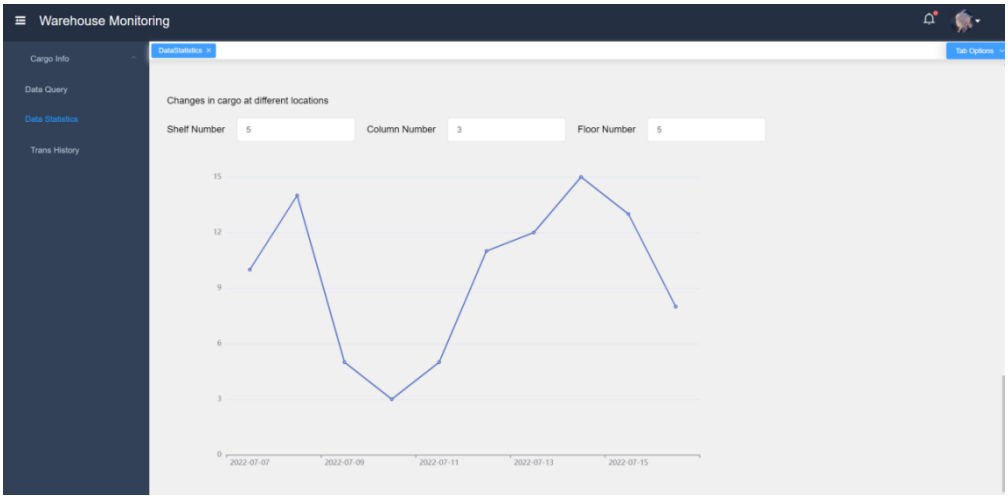
The following picture shows the realization of showing the change in the volume of a category of goods as time goes by.



Also, we can still check out a pie chart about the proportion of goods in different categories. The different 5 colors stand for disparate cargoes. It's definitely convenient for comparation and future plan about storing what kinds of goods.



Finally, the change in the quantity of goods at different locations can be seen through the line graph, if typing the shelf number, column number, as well as the floor number.



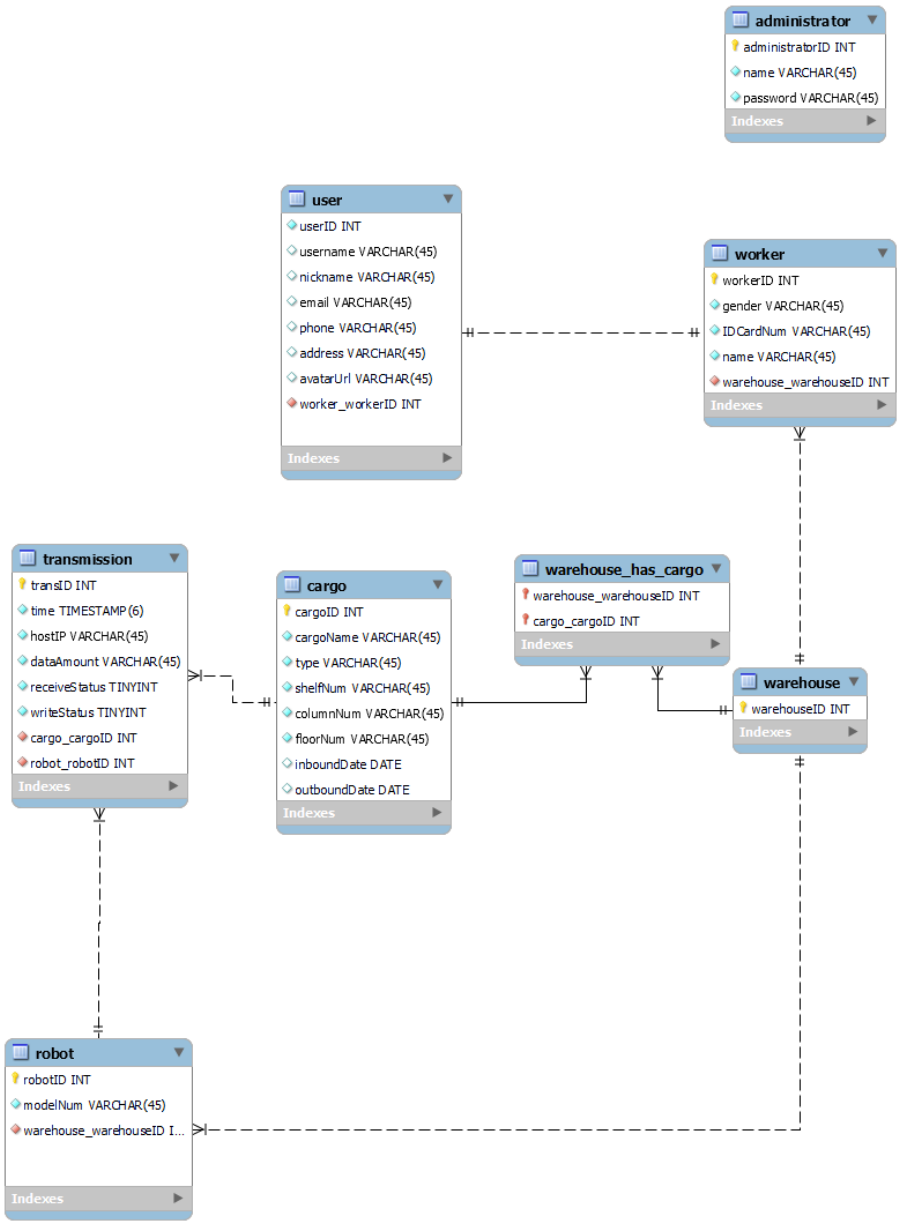
(13) Summary

To sum up, the webpage realizes multiply innovative functions:

- Data visualization by graph, chart, etc.
- Data shifting in real time transmitted by hardware.
- Have the same style of page, whether in login page or in warehouse monitoring page.
- User-friendly. Users can personalize their information, and for some elderly people or someone who has difficulty in remembering passwords, they can change their passwords quickly and use the page as normal.
- The layout is really fascinating and attractive, which can give users a comfortable experience. Also, the division of each module on webpage for workers and administrators is intuitive and clear.

2.2.3. Database

ER Diagram



ER Explanation

IDs are primary keys and increase automatically.

(IoT) user: each worker is a user of the warehouse supervision system,

- userID
- username
- password
- nickname
- email
- phone
- avatarUrl

- worker_workerID: foreign keys are used to associate workers

(IoT) cargo: cargoes in the warehouse

- cargoID
- cargoName
- type
- shelfNum: shelf number of cargoes
- columnNum: column number to which the cargoes belong
- floorNum: floor number of cargoes
- inboundDate
- outboundDate

(IoT) transmission: transmission history

- transID: unique identification transmission record
- time: transmission time
- hostIP: IP address of the transmitter
- dataAmount: data size
- receiveStatus: receiving status (1: success, 0: failure)
- writeStatus: write status (1: success, 0: failure)
- cargo_ CargoID: cargo ID associated with foreign key
- robot_ RobotID: the robot ID sent and transmitted by foreign key association

(IoT) robot: robot car

- robotID
- modelNum: robot model number
- warehouse_warehouseID: warehouse ID to which the robot belongs
- warehouse：(e-com) warehouse

(e-com) worker: worker's personal information

- workerID
- gender
- IDCardNUm
- name
- warehouse_warehouseID

(e-com) administrator: administrator’s personal information

- administratorID
- name
- password

3. Business Analysis

3.1. Design Concept and Advantages

- **Humanized design**

Our web page does not have a miscellaneous installation process, customers just need to visit the link to jump to our page. In addition, our web page also carries on the visualization technology management to the warehouse, which means real-time changes in the warehouse and shelf information changes can be reflected in our web page, so that it is convenient for users and managers to query.

- **Simplicity of operator**

Our website is aimed at all audiences, including the elderly who are weak in using computers, therefore the operation process needs to be concise and clear, so as to remove the hidden danger that some groups have difficulty in using the web page.

3.2. PESTEL Analysis

- **Political**

In recent years, China has issued "the Implementation Opinions on Internet Efficient Logistics", "the Opinions on Further Promoting the Cost Reduction and Efficiency Enhancement of Warehouse to Promote the Development of the Real Economy", "the Opinions on Promoting the Coordinated Development of E-commerce and Express Logistics", "the Implementation Plan for Promoting the Deep Integration and Innovative Development of the Logistics and Warehouse Manufacturing Industry" and other policies vigorously encourage and support the intelligent logistics equipment industry, with the goal of further reducing the total logistics cost of the whole society and improving the warehouse efficiency, providing a good policy environment for China's intelligent logistics and warehouse equipment industry.

- **Economic**

The number of newly registered enterprises in China's intelligent warehousing industry has increased from 101 in 2018 to 4015 in 2022. It is expected that the proportion will again increase in the future. The rapid development of China's logistics industry has provided a huge market demand for the rise of the warehousing industry. In addition to the release of outsourcing demand and policy guidance of the manufacturing industry and trade circulation industry, the strategic position of intelligent warehousing will continue to be strengthened. In the future, the demand for intelligent warehousing market will be further released. It is estimated that by 2023, the scale of the intelligent warehousing market will reach 197.5 billion yuan.

- **Sociocultural**

Intelligent warehouse is the product of warehouse automation and a link in the logistics process. From a social point of view, numerous industries in China have launched intelligent warehouse

management systems, such as auto parts, hardware, electronics, machinery, food, medicine and other industries. The application of intelligent warehouse ensures the speed and accuracy of data input in all links of cargo warehouse management, so that enterprises can timely and accurately master the real data of inventory, and reasonably maintain and control enterprise inventory.

Plus, in society, with the increase in land use costs and labor costs, storage costs have also increased significantly. Thus, the cost reduction and efficiency enhancement are the core of the future development of China's storage industry.

- **Technological**

The development of Internet of Things technology and 5G technology has realized the remote monitoring and maintenance of the logistics equipment system and improved the utilization rate and operation and maintenance efficiency of the intelligent transmission and sorting system. The industrial IoT network will soon become an essential part of efficient warehouse management and will provide the connections and data that intelligent warehouses rely on.

Big data can collect data in real time, improving the automation degree, operation efficiency and accuracy of intelligent transmission and sorting equipment, which is the development direction of future industry technology.

- **Environmental**

In recent years, China's intelligent warehousing scheme has enjoyed a strong momentum of going to sea, and the first echelon of logistics robot enterprises have made remarkable achievements. Typical countries and regions going to sea include the United States, Europe, Australia, Japan, South Korea, etc., which are mainly concentrated in areas with large demand for warehousing and logistics automation, such as high warehouse rent, commercial development, high labor costs, and low labor efficiency.

Therefore, China's cross-border e-commerce is in the window period of rapid development, and the demand for overseas warehouses is growing rapidly. The demand for warehousing automation market is huge.

- **Legal**

In recent years, our government and industry competent departments have attached great importance to the development of intelligent logistics and warehousing equipment industry.

The Ministry of Industry and Information Technology and other competent departments have successively issued a series of legal documents to vigorously support the development of the industry. For example, the Guidelines for the Construction of the National Intelligent Manufacturing Standard System, issued in August 2018, provides legal norms for intelligent logistic standards, mainly including intelligent warehousing standards such as material identification, material storage location allocation, operation status and equipment status control, and real-time cargo monitoring.

3.3. SWOT Analysis

- **Strength**

We are a newly established 10-person company with low development costs.

Our system is user-oriented and can distinctly show the goods in the warehouse on the web page, which is easy for customers to use.

Our warehouse management system simulates the real warehouse situation and is professionally designed. We utilize various sensors for goods identification, environmental monitoring, and emergency control.

- **Weakness**

The stability and creativity of our product still need to be improved.

At present, our system has only developed the web version, without expanding the function into application on cellphones for realizing better convenience.

- **Opportunity**

In the initial stage of entering the market, we can attract more users with a relatively low price.

In addition to updating and maintaining our system, we can also find investors and sponsors who are interested in our project.

We can combine with major supermarkets and apply the system in supermarket sales, so as to increase the number of customers.

- **Threat**

As the number of companies developing warehouse management systems increases year by year, our competitors increase, too.

To win more market share, we need to invest a lot of money and manpower to do the publicity, but it is a huge cost for our tiny company of ten people.

The increase in competitors means that we must be creative as soon as possible or we will lose market competitiveness.

3.4. Competitive Analysis

- **User Friendly View**

Our website is user-friendly. As a user, if you want to view the goods in the warehouse, you can log on the website directly, and follow the information prompt on the web page to operate. You can query the warehouse number, shelf number, price, and additional information about the goods you need easily, which is helpful for you to check every feature about the goods.

- **Maintainability View**

Our company's website can update and iterate information in real time. If you encounter difficulties in using the website, you can call the consultation number to provide advice or seek help for our company's products. Also, the hardware to detect the goods is stable and durable.

- **Prospect View**

The tenet of our company is to put customers first. We will uphold the tenet of our company and provide customers with as much help as possible.

3.5. 4P Strategy

- **Product**

Our product is the application of an intelligent warehouse management system, which ensures

the speed and accuracy of data input in all links of cargo warehouse management, so that enterprises can timely and accurately master the real data of inventory, and reasonably maintain and control cargo warehousing.

- **Price**

When the enterprise is small and the warehouse area is around 2000 square meters, a customized simple warehousing system can be used. The price of such a warehousing system is commonly tens of thousands to hundreds of thousands of yuan, which can realize the systematic and information-based management of warehouse.

When the enterprise expands to medium scale or above, the price of a customized system usually needs between 100000 and 500000, depending on the functional category, difficulty, development workload, pricing, and positioning of service providers, etc.

- **Promotion**

Set up offline experience stores to attract more customers to experience our warehousing system on the spot.

Cooperate with e-commerce websites to insert our product links on the website interface.

Offer discounts to customers who purchase products early in the product launch.

Provide VIP service for users who have use our warehousing system for a long time.

Multiple payment methods are supported. Users can pay by installment or with credit card.

Provide after-sales services, such as door-to-door delivery, free installation, free warranty service and return service within a certain period.

- **Place**

First of all, we start our business in the first-tier cities with developed logistics and e-commerce industries. Then we expand the business, establishing subsystems in other cities.

4. Meeting Management

4.1. The Timeline

Electronic Engineering and Management

During the D&B class

8.22 Attend the class to meet with group members and understand project requirements roughly. After taking lessons from professors, group discussion is built to search the detailed information, then learn about Arduino, such as programming language and theory of each sensor. Set clear goals for each day and start assembling the infrastructure of the car.

8.23 Continue to learn about Arduino, implement some of the functions, such as sensing the temperature and humidity, recognizing colors and so on. Achieve initial integration of codes.

8.24 Realize all functions and continue to optimize them, making sure that each function can be well-operated independently.

8.25 Combine all functions and attach them to the hardware together, especially the tracking function. Begin to test and debug. Design a good look for the hardware.

8.26 The Chinese teacher check our car, seeing how it works. Hand in our source codes, video, and slides.

After the summer vacation

9.2 ~ 9.3 Continue learning about Arduino. Purchase innovative components to improve the realization of functions, as well as the steadiness of manufacturing. Email our progress to our Chinese teacher.

10.8 ~ 10.14 Get in touch with teammates in other majors. Ensure the successful operation of all functions in the hardware and enable wireless data transmission using Wi-Fi.

10.15 ~ 10.20 Realize sending information to website in real time. Continue to debug the code and improve the manufacturing. Complete the final report and hand in all materials we have done.

E-commerce and Law

7.3 Learn relevant theoretical knowledge of html, css and java web.

7.4 Learn to build a simple web project without connecting to the database

7.5 Learn to build a simple web project that connects databases. Design the database together with the IoT major. On this basis, start to realize login, logout, and query functions.

7.6 Realize warehouse information addition, deletion, and modification functions.

7.7 Integrate the code to create a simple website, and the Chinese teacher carried out the mid-term acceptance.

7.8-9.20 Improve site functionality and user experience, improve site quality.

9.20-10.20 Start integrating the components together and produce a complete system.

IoT

7.3 Discuss what we should do, and elementarily allocate task.

7.4 Decide the EER model.

7.5 Design the relationship.

7.6 Partly decide the port.

7.7~7.20 Start coding.

7.23~9.1 Start to design the website.

9.1~9.10 Do trimming to our code.

9.11~10.20 Combine our result with members in other majors.

4.2. Minutes of Meeting

Electronic Engineering and Management

- **First Meeting**

Date/Time: 22nd, August 2022, 4:00 p.m.

Location: S1 Teaching Building

Present: All students in our group

Leader: Tangxun Wu

Proceeding:

1. List all the tasks that should be done recently, including programs and hardware.
2. Divide work for individuals and make a clear timeline.
3. Check every component after distribution and see if everything can work normally.

- **Second Meeting**

Date/Time: 25th, August 2022, 9:30 a.m.
Location: S1 Teaching Building
Present: All students in our group
Leader: Tangxun Wu
Proceeding:
1. Assemble the hardware to form the car and decorate it.
2. Test the codes and see if there is any improvement that can be done.

- **Third Meeting**

Date/Time: 3rd, September 2022, 3:00 p.m.
Location: WIN basement of BUPT
Present: All students in our group
Leader: Tangxun Wu
Proceeding:
1. Improve the basic functions, especially the tracking function.
2. Determine what should be further accomplished.

E-commerce and Law

- **First Meeting**

Data/Time：3rd July 2022, 4:00 pm
Meeting form: online meeting
Present: all e-commerce major students
Major Leader: Jingru Dai
Proceeding:
1. Unified development environment.
2. Define tasks for each phase.

- **Second Meeting**

Data/Time：4th July 2022, 4:00 pm
Meeting form: online meeting
Present: all e-commerce major students
Major Leader: Jingru Dai
Proceeding:
1. Define the functions to be realized.
2. Clarify the division of labor and assign the functions to be realized by everyone.

- **Third Meeting**

Data/Time：5th July 2022, 4:00 pm
Meeting form: online meeting
Present: all e-commerce major students
Major Leader: Jingru Dai
Proceeding:
1. Discuss the entity and attribute necessary for the database.
2. Discuss the problems encountered in the realization of functions.

- **Final Meeting**

Data/Time：6th July 2022, 4:00 pm
Meeting form: online meeting
Present: all e-commerce major students
Major Leader: Xianglong Qin
Proceeding:
1. Introduce the final code, explain the principle of the code and the knowledge used.
2. Prepare for the interim acceptance.

IoT

- **First Meeting**

Time: 3rd July 2022, 1:00 pm
Meeting form: Zoom
Present: All IoT students
Leaders: Yifan Yang
Proceeding:
1. Discuss what should we do.
2. Allocate the task into pieces.

- **Second Meeting**

Time: 5th July 2022, 9:00 am
Meeting form: Zoom
Present: All IoT students
Leader: Zhuoheng Han
Proceeding:
1. Discuss what kind of data should we provide and what we will accept.
2. Discuss the function realization.
3. Solve the problems in coding.

- **Third Meeting**

Time: 8th July 2022, 9:00 am
Meeting form: Zoom

Present: All IoT students
Leaders: Yifan Yang
Proceeding:
1. Discuss the relationship between variables.
2. Decide the relation between webpages.

- **Forth Meeting**

Time: 10th July 2022, 9:00 am
Meeting form: Zoom
Present: All IoT Students
Leaders: Zhuoheng Han
Proceeding:
1. Run the entirety code.

Overall

- **First Meeting**

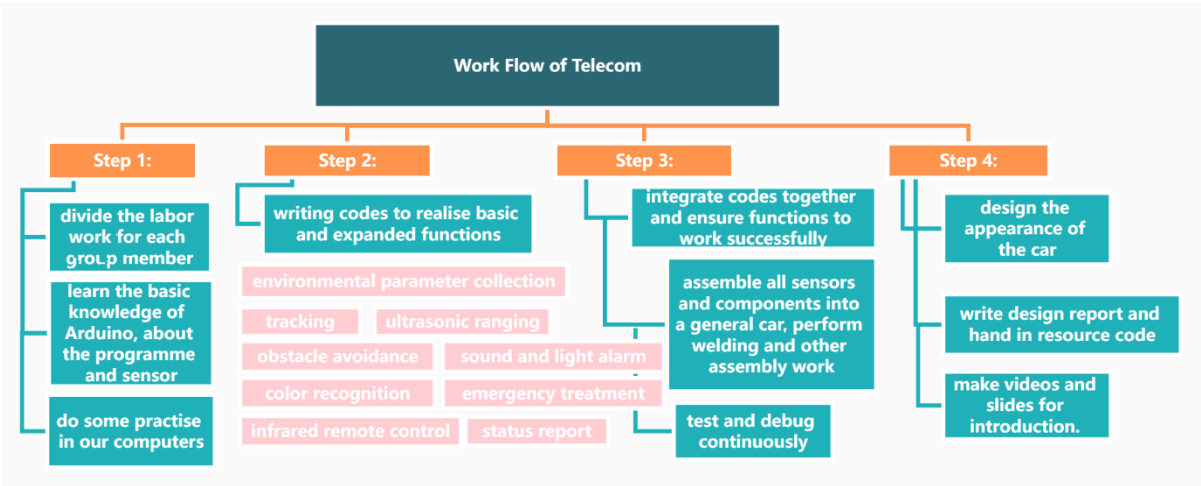
Date/Time: 7th, October 2022, 4:00 p.m.
Location: WIN basement of BUPT
Present: All students in 3 groups
Leader: Tangxun Wu
Proceeding:
1. Every group report what have done and what should be done.
2. Listing corporate works, like program docking, advertisement making.

- **Second Meeting**

Date/Time: 16th, October 2022, 4:00 p.m.
Location: WIN basement of BUPT
Present: All students in 3 groups
Leader: Tangxun Wu
Proceeding:
1. Finish program docking among 3 groups and test the achievement.
2. Allocate the tasks like report writing, appearance designing and program improving to fully prepare for the final check.

5. Workflow

Electronic Engineering and Management:

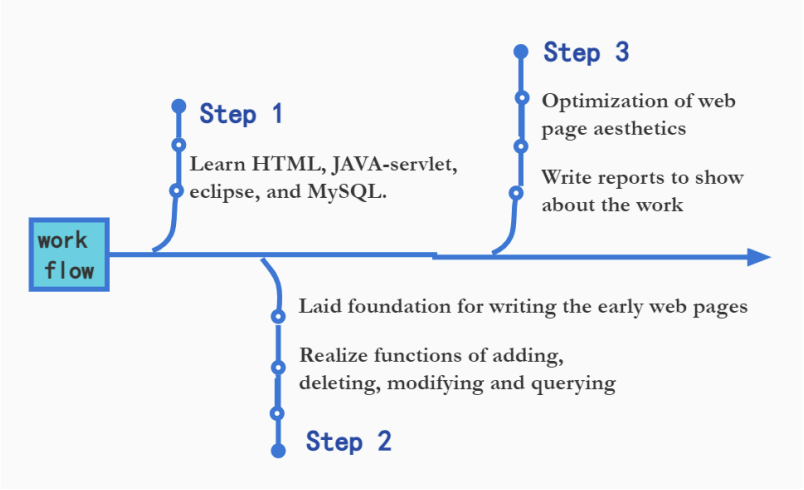


E-Commerce and Law:

Step 1: In the early stage of the project, we learned HTML, JAVA-servlet, eclipse, and MySQL.

Step 2: We laid a foundation for the writing of early web pages and used that software to write early versions of web pages, such as the functions of adding, deleting, modifying, and querying warehouse shelves. SQL language (workbench) was used to compile the database.

Step 3: In the later stage, under the premise of realizing the basic functions, the optimization of web page aesthetics.



IoT:

Step1: Learn SQL, Spring frame, and even Go (at first we want to use this, but give up later)

Step2: Decide the option we should do

Step3: Initially complete

Step4: Page optimization

6. Division

- **Hardware (Electronic Engineering and Management)**

Individual Member 1 Contribution: <Tangxun Wu (Leader)>

Software and hardware realization of temperature and humidity sensor, color sensor, flame sensor; Implementation of infrared obstacle avoidance system; Assist to assemble hardware and code modification debugging; Responsible for hardware appearance and PPT modification; Adjust the working process and assign tasks to group members in three majors; Complete the draft of advertisement and integration of report.

Individual Member 2 Contribution: <Boyang He>

Realizes the ultrasonic ranging module and infrared remote-control module; Integrate ultrasonic and temperature and humidity sensor modules; Use LCD to display information about color, distance, location, humidity, and temperature; Integrate all the sensors into hardware, and connect the codes; Responsible for wireless communication module, transmitting data to the webpage.

Individual Member 3 Contribution: <Dongze Sun>

Software and hardware implementation of the photosensitive module; Combination of photosensitive module and buzzer to realize the over-brightness alarm; Test the vibration module; Debug the driver module and the Bluetooth module; Test the infrared remote control and tracking module.

Individual Member 4 Contribution: <Weiping Kong>

Perform circuit welding and wiring optimization; Debug infrared tracking; Test collision sensor; Debug Bluetooth module to achieve information transmission; Test infrared remote control and tracking module; Appearance design and manufacture; Shoot video materials.

Individual Member 5 Contribution: <Guanyi Lin>

Online using simulation tools to complete ranging and other code writing and wiring; Consult and share relevant data of Bluetooth, infrared remote control, LCD, and buzzer, and assist offline team members in their work; Complete the first version of the PPT report; Make the video for presentation.

- **The Website Application (e-Commerce)**

The e-commerce major is mainly responsible for designing databases with IoT students and creating web applications with the following functions:

1. Allow administrators to log in and log out.
2. Allows administrators to create warehouses and add/delete workers in the system.
3. Display the goods status (such as goods location) and information (such as goods type) of each

warehouse.

Individual Member 1 Contribution: <Jingru Dai>

Design the overall process and framework. Realize login function and employee information query function. Website front-end jsp: login page, management page, worker query page, worker information page. Unified beautification of the overall page.

Individual Member 2 Contribution: <Xinyi Li>

Realize logout function and warehouse information query function. Website front-end jsp: warehouse query page, warehouse information page, and warehouse creation page. Unified beautification of the overall page.

Individual Member 3 Contribution: <Xianglong Qin>

Realize the functions of goods information query and warehouse information addition, deletion, and modification. Create the database. Write the XML file. Website front-end jsp: goods query page, goods information page, warehouse information addition/deletion/modification page. Integration code.

- **Internet of Things**

Individual Member 1 Contribution: <Yifan Yang>

Create the database, the development of work interface, the transmission history part and the cargo query.

Individual Member 2 Contribution: <Zhuoheng Han>

The design of EER model, log-in interface part, the admin interface part.